

IUL
Islamic University Of Lebanon
Université Islamique Du Liban
الجامعة الإسلامية في لبنان

StudyNexus: AI-Powered Study Planner

By

Malak Yazbek

Graduation Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Computer Science

Department of Computer Science

Faculty of Sciences & Arts

2025 – 2026



IUL
Islamic University Of Lebanon
Université Islamique Du Liban
الجامعة الإسلامية في لبنان

StudyNexus: AI-Powered Study Planner

By

Malak Yazbek

Graduation Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Computer Science

Department of Computer Science

Faculty of Sciences & Arts

Supervised by

Dr. Mokhaiber Dandash

Department of Computer Science

2025 – 2026



IUL
Islamic University Of Lebanon
Université Islamique Du Liban
الجامعة الإسلامية في لبنان

The Report Defense Committee for *Malak Yazbek* Certifies that this is the approved version of the following report

StudyNexus: AI-Powered Study Planner

APPROVED BY :

Supervisor Signature : _____

Examiner Signature : _____

Examiner Signature : _____

Acknowledgments

I would like to say thank you, sincerely to **Dean Dr. Ali Raad** for his remarkable leadership, his continuous encouragement, and the dedication he has shown in shaping an academic atmosphere that supports excellence, creativity and innovation. I am also really grateful to **Dr. Samar Chamas** for her useful guidance, for her insightful feedback, and for her constant support all along my academic path. Her commitment and encouragement have been genuinely meaningful, and I truly appreciate it.

My deepest gratitude goes to my project supervisor, **Dr. Mokhaiber Dandash**, for his guidance, patience, helpful feedback, and continuous support throughout this project. His knowledge, advice, and encouragement helped me solve problems, improve my work, and always do my best. I truly appreciate the time and effort he gave to support me. Without his guidance and support, this project would not have been completed successfully.

This project is really the result of months of dedication, perseverance, and ongoing learning. The development of StudyNexus: AI-Powered Study Planner required extensive research, careful organization, persistent problem solving, and lots of hours spent on design, implementation, testing and refinement. Every time something unexpected came up during development, it turned into a useful learning experience that improved my technical knowledge, strengthened my analytical ability, and sharpened my problem-solving skills.

And finally, I want to express my heartfelt appreciation to my family, and to everyone who supported me and encouraged me during this journey. Their patience, their motivation, and their unwavering confidence in my capabilities gave me the strength, and the determination I needed to pass through the difficulties and finish this project successfully.

Abstract

StudyNexus: AI-Powered Study Planner

The rapid growth of digital technology in education has changed the way students plan and manage their academic work. However, many students still face problems such as poor planning, unorganized tasks, limited progress tracking, and low motivation. Traditional study management methods often do not provide personalized suggestions or real-time analysis of student learning habits. To solve these problems, this project introduces **StudyNexus**, an AI-powered cross-platform study planning system designed to improve academic organization, productivity, and student engagement.

StudyNexus was developed using Flutter for the mobile application, Firebase for real-time cloud services and data storage, and a **PHP-based admin website** for system management. The system uses **Gemini 2.5 Flash** to provide smart study suggestions, automatic planning, and personalized learning insights. It includes many productivity features, such as a planner with daily, weekly, and monthly views, a Pomodoro study timer, and an AI-powered weekly scheduling system that creates study plans based on deadlines, exams, available time, and student priorities.

The system also includes performance tracking features that analyze student study habits using measurements such as focus score, productivity score, and study streaks. A time-weighted average method is used to calculate performance scores so that longer and more productive study sessions have a greater effect on the final results. In addition, the AI Insights feature analyzes past study data to identify study patterns, estimate learning efficiency, detect possible signs of burnout, and provide personalized suggestions for improving study habits.

Besides helping students manage their studies, StudyNexus also includes social features that encourage collaboration. These features include a friend recommendation system based on mutual friends, academic major, and study level, as well as real-time messaging and weekly leaderboards that increase motivation and engagement. The admin website also provides tools for user management, AI usage monitoring, and an anti-cheat system to help ensure that recorded study activities are reliable.

The completed system shows that combining artificial intelligence, cloud technology, and study behavior analysis can create a modern and effective educational platform. By providing personalized planning, data-based insights, progress tracking, and professional PDF study plan generation, StudyNexus helps students improve time management and academic performance. This project demonstrates how AI-powered educational applications can support more personalized and effective learning in today's digital environment.

Table of Contents

General Introduction 1
Summary of Chapters Included in the Report 2
Chapter 1: System Planning 4
1. What StudyNexus Actually Is 4
2. Aim and Objectives 4
2.1 The Main Aim 4
2.2 Five Core Objectives 5
3. Project Scope 6
3.1 The Student Experience 6
3.2 Platform Supervision and AI Governance 6
4. Development Environment 7
4.1 Languages, Frameworks, and Tools 7
4.2 Database and Cloud Services 7
5. Why This Matters 8
Chapter 2: Related Work 9
1. Purpose of the Review 9
2. Existing Approaches, by Category 9
2.1 Workspace & Task-Management Apps 9
2.2 Dedicated Academic Planners 10
2.3 Focus & Gamification Timers 10
2.4 AI-Based Calendar Automation 10
2.5 Research Projects & Academic Studies 11
3. Comparison Table 11
4. How AI is Used in StudyNexus 13
5. Technology Used 14
6. Summary of Gaps & StudyNexus's Position 15
Chapter 3: Requirements Analysis 16
1. Overview 16
2. Requirements Gathering 16
3. User Groups 16
4. Functional Requirements 17

5. Non-Functional Requirements 18
6. User Needs 18
7. Use Case Diagram 19
8. Data Flow Diagrams (DFD Level 0 and 1) 21
9. Class Diagram 22
10. Summary 24
Chapter 4: System Design 25
1. Introduction 25
2. Database Design 25
3. Data Relationships and Access Patterns 26
4. System Architecture 26
5. Summary 27
Chapter 5: System Implementation 28
1. Introduction 28
2. Development Environment and Technology Stack 28
3. Database Implementation 29
4. User Account Management 30
5. Mobile Application Implementation 31
6. Artificial Intelligence and Analytics Implementation 33
7. Web-Based Administration Panel Implementation 33
8. Data Validation and Security 34
9. Summary 34
Chapter 6: Testing and Results 35
1. Website Testing 35
2. Mobile App Testing 49
General Conclusion and Future Recommendations 74
References 76

List of Tables

Table 1: Technologies Used in StudyNexus 7

Table 2: Comparison of StudyNexus with Existing Study and Productivity Applications 12

Table 3: StudyNexus System Modules and Their Descriptions 18

Table 4: High-Level Firestore Database Schema of the StudyNexus Application, Representing the Main Collections and Their Functional Responsibilities 26

Table 5: Overview of the Firestore Database Collections, Data Organization, and Purpose within the StudyNexus System 30

Table 6: Description of StudyNexus Application Views and Their Functional Purposes 31

List of Figures

Figure 1: StudyNexus UML Use Case Diagram illustrating primary user roles (Registered Student, Guest User, and Administrator) and interactions with Firebase services and Gemini AI 20

Figure 2: Level 0 Data Flow Diagram (Context Diagram) of the StudyNexus System 21

Figure 3: Level 1 Data Flow Diagram (DFD) illustrating the interaction between StudyNexus system processes, external entities, and data stores 22

Figure 4: UML Class Diagram Illustrating the Core Entities, Attributes, Relationships, and Data Models of the StudyNexus System Architecture 23

Figure 5: Landing page of the StudyNexus website, introducing the platform's AI-powered study intelligence 35

Figure 6: StudyNexus password recovery interface with success and error toast notifications 36

Figure 7: StudyNexus Admin Login Interface for administrator authentication with toast messages 36

Figure 8: StudyNexus Admin Dashboard showing the main administrative modules and system management features 37

Figure 9: StudyNexus user profile management interface displaying account details, avatar settings, and profile options 38

Figure 10: Edit Profile screen interface showing user information fields and account update options 38

Figure 11: StudyNexus Change Avatar interface flow, showing avatar upload, preview, and success/error states 39

Figure 12: User, Guest, and Admin Records dashboard interface, featuring search, filtering options, user information table, and management actions 40

Figure 13: Delete User confirmation interface with user details and permanent deletion action 41

Figure 14: Activity Logs interface showing user management, activity tracking, and failure message notifications 42

Figure 15: Compose Admin Message interface for creating and sending failure notifications to users 43

Figure 16: StudyNexus AI Control Center interface with AI schedule and insight management panels 43

Figure 17: StudyNexus AI Schedules Control Page displaying AI prompt configurations, generated schedules, and overall system statistics 45

Figure 18: StudyNexus AI Insights Control Page displaying AI prompt configurations, generated insights, and overall system statistics 46

Figure 19: StudyNexus Motivation System dashboard displaying user statistics, leaderboard rankings, streak tracking, and admin notifications 47

Figure 20: Activity validation and warning history sections showing detected issues and administrator warning records 48

Figure 21: StudyNexus authentication interface, providing user login, registration, and guest access options within a unified page 49
Figure 22: StudyNexus interface states showing alerts, messages, and account management dialogs 50
Figure 23: StudyNexus password reset interface, illustrating validation states and Firebase recovery email flow 51
Figure 24: StudyNexus Home Dashboard management interface, featuring user data synchronization, study metrics, notifications, weekly study hours tracking, and friend leaderboard rankings 51
Figure 25: NexusConnect interface displaying user profiles, friends, friend requests, and user suggestions 52
Figure 26: User profile search interface with friend request management 53
Figure 27: NexusMessages messaging interface showing conversation list, empty chat, and active conversation screens 54
Figure 28: Chat module showing an empty state and a friend with unread messages 54
Figure 29: Study subjects management interface showing the list, forms, and dialog screens 55
Figure 30: StudyNexus task management interface, featuring the task dashboard with priority and status filters and the task creation form 56
Figure 31: StudyNexus task management interface showing the task dashboard with filtering options, task details, and task editing screens 57
Figure 32: StudyNexus planner interface showcasing day, week, and month calendar views, schedule creation, and validation toasts 58
Figure 33: StudyNexus smart planner interface showcasing AI-generated schedules and PDF schedule exports 59
Figure 34: Study Session interface showing session setup, active timer, session termination, and successful save confirmation 61
Figure 35: Session reflection interface with optional task completion tracking 63
Figure 36: StudyNexus AI insights dashboard analyzing study habits, performance trends, and personalized recommendations 63
Figure 37: NexusAnalytics dashboard interfaces displaying study tracking metrics, including consistency, subject performance, and learning trend analysis 65
Figure 38: StudyNexus analytics page displaying study patterns, time distribution, productivity metrics, and focus behavior insights 66
Figure 39: User profile and account management interfaces in the StudyNexus application, comparing guest and registered user accounts, including sign-up with guest account linking, profile editing, password updates, and avatar customization 69
Figure 40: NexusInbox notification interface showing category filters and notification management features, including read, unread, and delete actions 71
Figure 41: StudyNexus push notification interface displaying schedule reminders for upcoming and active study sessions 73

General Introduction

The fast growth of digital technology has changed the way students access information, manage their responsibilities, and organize their daily study activities. Although many digital tools are available today, many students still have difficulties keeping a good study routine, organizing academic tasks, and tracking their progress. Many students still use traditional methods such as handwritten notes, paper planners, or different applications to manage their studies. These methods can become difficult to manage over time, making it harder for students to stay organized, set priorities, and remain motivated.

The lack of one complete and smart solution creates a challenge for students who need an easy way to manage their academic work while using modern technology. Creating study schedules, organizing tasks, and checking progress manually takes time and effort, which can reduce students' motivation to continue planning their studies. Therefore, there is a need for a digital platform that helps students organize their academic life, provides personalized support, and encourages better study habits through an easy and engaging user experience.

This project introduces StudyNexus, an AI-powered study planner designed to help students manage their academic activities in an organized and motivating way. The system includes smart planning features such as automatic weekly schedule generation using the Gemini Large Language Model (LLM), task and subject management, study session reflection, and study performance analysis, including focus tracking and fatigue detection. To increase motivation and encourage regular studying, the platform also includes a friends' leaderboard that ranks students based on their weekly study minutes.

The development of StudyNexus follows the main stages of software development, including system planning, requirements analysis, system design, implementation, and testing. The system consists of a mobile application for students developed using Flutter and an administrative web portal developed using PHP. Instead of using a separate custom API, the PHP web portal connects directly to Firebase through the secure Firebase REST API using OAuth 2.0 server-to-server authentication. Firebase provides the backend services, including Cloud Firestore for data storage, real-time user presence synchronization, and secure Email/Password authentication. This combination of technologies provides a secure, reliable, and responsive platform that supports both student learning and administrative management.

Summary of Chapters Included in the Report

Chapter One: System Planning

This chapter introduces StudyNexus, an AI-powered study planner created to help students solve common problems such as poor time management, low motivation, and difficulty organizing their studies. It explains the project's goals, objectives, scope, feasibility, and development method. It also describes how the Agile SDLC was used to support step-by-step development and regular improvements.

Chapter Two: Related Work

This chapter reviews existing study planning applications, educational technologies, and previous research on artificial intelligence in education. It explains the use of Large Language Models (LLMs), gamification, and social learning. It also compares existing solutions to show the problems that StudyNexus aims to solve with smart scheduling, learning analytics, and AI-based recommendations.

Chapter Three: Requirements Analysis

This chapter describes the functional and non-functional requirements of the system. It explains the needs of the users, the main system features, performance and security requirements, and the technical limitations that were considered during the design and development of StudyNexus.

Chapter Four: System Design

This chapter explains the overall design of the system, the Firestore NoSQL database structure, and the user interface. It describes how the Flutter mobile application and the PHP admin website work with Firebase services to provide a simple and responsive user experience.

Chapter Five: Implementation

This chapter explains how the StudyNexus system was developed. It describes the Flutter mobile application, the behavior analysis system, the Gemini AI integration, and the PHP admin dashboard. It also explains the implementation of smart scheduling, AI study insights, user authentication, and system management features.

Chapter Six: Testing and Results

This chapter presents the testing process used to check the functionality, reliability, and performance of the system. It evaluates the API performance, database operations, analytics accuracy, caching, and the overall system performance. The results show that StudyNexus is an effective, reliable, and scalable solution for academic study planning.

The report ends by presenting possible future improvements that can help enhance and expand the StudyNexus system.

Chapter 1: System Planning

1. What StudyNexus Actually Is

StudyNexus consists of two connected systems that work together. The first system is a Flutter mobile application that students use every day to organize their study time, manage tasks, record study sessions, chat with friends, and receive AI-based study suggestions. The second system is a PHP-based admin website that allows administrators to manage users, monitor the system, and oversee the AI features used by the application.

Both systems use Google Firebase as the main backend service. Cloud Firestore stores important data such as user profiles, subjects, tasks, schedules, study sessions, messages, notifications, and friendships. Firebase Realtime Database is used to show users' online status in real time. The admin website does not connect to Firestore using the Firebase SDK. Instead, it uses OAuth 2.0 authentication with a service account and communicates with Firestore through its REST API. This design allows the mobile app and the admin website to work independently while using the same database.

One of the main features of StudyNexus is the integration of Google's Gemini 2.5 Flash AI model. Instead of only answering questions, Gemini helps students by creating personalized weekly study schedules and providing useful study recommendations. It also analyzes the student's study behavior to generate insights such as burnout risk, focus suggestions, and performance trends, helping students improve their learning habits.

2. Aim and Objectives

2.1 The Main Aim

The project aims to improve students' productivity and study habits by combining two features that are usually separate: task and schedule management, and AI-based personalization. It provides a mobile application for students to organize their studies and a secure admin website for administrators to manage the platform and monitor the AI system.

This two-sided design is an important part of the project. Many study apps either leave everything to the student, requiring them to stay motivated on their own, or depend completely

on AI without explaining its decisions. StudyNexus takes a balanced approach. The AI creates study schedules and gives personalized suggestions, while students can view and change these recommendations if needed. At the same time, administrators can review the AI's outputs and the information used to generate them. This connection between the student, the AI, and the administrator is an important part of the project's objectives.

2.2 Five Core Objectives

- AI-Powered Scheduling & Insight Engine — Gemini 2.5 Flash generates weekly schedules from tasks, exams, free time, and recent behavior, while enforcing real scheduling constraints: protected sleep windows, 30–90 minute study blocks, deadline-awareness, and spaced repetition. The same pipeline produces cache-aware insights validated against a strict JSON schema before anything is stored.

- Cross-Platform Mobile Application — A Flutter app covering authentication, task and subject management, day/week/month planning, Pomodoro and manual timers, friends and messaging, notifications, AI tools, and analytics, all synced in real time with Firestore.

- Secure Administrative Web Portal — A PHP portal with session-based access control and CSRF-protected forms, talking to Firestore over an authenticated REST API, giving admins the ability to manage accounts, monitor activity, configure Gemini prompts, and supervise AI output.

- Real-Time Sync & Presence Tracking — Real-time listeners and lightweight heartbeat mechanisms, running on both the Flutter client and PHP backend, keep profiles, presence, notifications, schedules, and messages in sync across devices without driving up Firestore read/write costs.

- Analytics & Gamification for Consistency — Streak tracking, a weekly friends leaderboard, study-minute-based unlockable avatars, and the 'Nexus Analytics' dashboard (consistency heatmaps, subject intelligence, behavioral analytics, time-efficiency insights) turn raw study data into motivation.

3. Project Scope

3.1 The Student Experience

Everything a student can do lives inside ten dashboard modules reachable after login: Home, Friends, Messages, Subjects, Tasks, Planner, Study Sessions, AI Tools, Performance Analytics, and My Space. The journey starts with registration or login — including guest accounts that can be created instantly and upgraded to a full account later — and extends through profile and avatar customization, subject and task management, AI-assisted or manual scheduling, a Pomodoro-enabled session timer with break tracking, post-session reflection, and social features like friend management and direct messaging.

3.2 Platform Supervision and AI Governance

The admin side is designed to focus on trust and control instead of having many different features. Only users with the **admin** role can access the portal. It allows administrators to manage user accounts by searching, filtering, enabling, disabling, or deleting accounts. They can also view user activity logs and send messages directly to students through the app.

Two features are the main parts of this system. The first is the motivation and leaderboard management dashboard, which includes an anti-cheat system. This system checks study records and flags suspicious sessions. For example, if a student marks a study session as completed in a very short time that does not match the amount of work recorded, the session is marked for review.

The second and most important feature is the AI management system. StudyNexus includes two AI dashboards: one for managing AI-generated study schedules and another for managing AI-generated academic insights. These dashboards allow administrators to edit the AI system prompt and review previous AI responses for each student. Every change made to the prompt is recorded, so administrators can see who made the change and when.

This management system makes the AI more transparent instead of acting like a hidden system. Administrators can review, correct, and monitor the AI's decisions. This is especially important because the AI gives recommendations that can affect students' study plans, workload, and wellbeing.

4. Development Environment

4.1 Languages, Frameworks, and Tools

Technology	Role in StudyNexus
Dart & Flutter	Used to build the mobile application for Android and iOS, including the user interface and app management.
PHP	Used for the admin portal backend, including login security, protection against unauthorized requests, and connecting with Firebase Firestore.
JavaScript / HTML / CSS	Used to create interactive admin pages, including searching, filtering, sorting, and loading data without refreshing the page.
Gemini API (2.5 Flash)	Used to generate AI study schedules and personalized academic recommendations with a fixed JSON response format.
Firebase Authentication SDK	Used for user login and account management in the mobile application, including email/password login and guest accounts.
REST API / OAuth 2.0 (JWT)	Used to securely connect the PHP admin portal with Firebase Firestore and allow the server to access stored data.

Table 1: Technologies Used in StudyNexus

4.2 Database and Cloud Services

- Cloud Firestore — primary NoSQL database: users, subjects, tasks, schedules, study sessions, messages, notifications, friendships, AI configuration, and AI-generated insight/summary collections.
- Firebase Authentication — manages user identities and login sessions for the mobile app.
- Firebase Realtime Database — lightweight, high-frequency data such as online-presence status and the guest-account naming counter.
- Firestore REST API with service-account OAuth — lets the PHP admin portal securely read and partially update Firestore documents server-side.

- Gemini API — the shared AI service consumed by both the mobile client (schedules, insights) and the PHP backend (prompt configuration and supervision).

5. Why This Matters

This chapter shows that StudyNexus isn't just another to-do list with an AI tacked on. It's a real productivity system, where AI isn't a gimmick — it's built right into the core, and it's managed from the start. Three big design choices set the tone for everything that follows:

• Personalization is rooted in real data. Gemini doesn't just spit out canned responses — it pulls from actual tasks, sessions, and real behavioral analytics. Every suggestion runs through JSON-schema validation and caching before getting to a student or landing in an audit log.

• Governance is baked in from day one. The system isn't scrambling to play catch-up with oversight after the fact. Dual supervision dashboards and a permanent audit log let admins trace, fix, and control the exact prompts guiding student recommendations.

• Real-time performance is balanced with cost. The heartbeat-and-listener method for tracking presence and syncing data isn't just for flash — it's designed to feel fast to the user, but without running up Firestore costs.

The rest of this report builds on that foundation. After these system-planning decisions, the focus shifts to analyzing, designing, and building both the student app and the admin portal that oversees it.

Chapter 2: Related Work

1. Purpose of the Review

Let's look at how students manage their time, track their focus, and use AI to plan their studies—and where StudyNexus fits in. Students have a lot to balance: deadlines pile up, some classes are way tougher than others, and study routines tend to fall apart. Plenty of digital tools try to help, but most only tackle one slice of the problem. You'll find one app for to-do lists, another for scheduling, and maybe a game-like tool that tries to keep you focused. What's missing is something that pulls it all together—a system linking scheduling, behavior tracking, and keeping students motivated.

This chapter is split into four sections. First, there's a look at current study and productivity apps, sorted by how they actually approach the problem—including some interesting research projects. Then, the focus shifts to how AI and machine learning get used for personalized schedules and adaptive learning. After that, there's a dive into the mobile and web tech behind these tools, showing where StudyNexus's own stack—Flutter, PHP, and Firebase—fits in. Finally, the chapter wraps up with a rundown of what most tools still lack, which is right where StudyNexus steps in.

2. Existing Approaches, by Category

Digital study-planning tools can be divided into four main groups: general workspace and task-management apps used for studying, academic planning apps, focus and motivation timer apps, and AI-based calendar planning tools. Each group helps students improve productivity in a different way, but each one also has limitations. These limitations help explain the need for developing StudyNexus.

2.1 Workspace & Task-Management Apps

Notion (+ AI) is a flexible workspace that many students use to create their own study systems. Students can create databases to organize assignments, reading materials, and study schedules. Its main advantage is its flexibility because it allows users to connect notes, deadlines, and progress tracking in one place. Notion AI can also help summarize notes and create study plans.

However, this flexibility is also a weakness for students who need a ready-to-use study planner. Notion does not provide a built-in academic structure, and students usually need to spend a lot of time setting it up or using templates made by others before it becomes useful [1].

Todoist / TickTick: These applications focus on simple task management by providing to-do lists with features such as due dates, repeated tasks, and priority levels. They are useful for students who only need to track their tasks and deadlines. However, they are not designed specifically for academic needs, such as managing subjects, semesters, or study sessions. They also do not analyze student performance, meaning they cannot update study plans based on missed sessions, focus levels, or subject difficulty [2], [3].

2.2 Dedicated Academic Planners

Applications such as **MyStudyLife** and **StudySmarter** are designed specifically for students. They provide features like class schedules, exam countdowns, flashcards, and tracking of course progress. These tools are better than general task-management apps because they understand the structure of a semester. However, their analysis is mostly limited to showing completed work. They do not deeply analyze study habits, measure study effectiveness, or automatically adjust plans when students miss study sessions [4], [5].

2.3 Focus & Gamification Timers

Apps such as **Forest** and **Study Bunny** focus mainly on helping students stay concentrated during study time rather than planning their academic tasks. Forest creates a virtual tree that grows while students avoid using their phones, while Study Bunny rewards students with coins for completing Pomodoro study sessions. These apps help increase motivation and the number of completed study sessions through game elements. However, they are not connected to student deadlines, subjects, or academic progress, which means the motivation system is separate from actual study goals [6], [7].

2.4 AI-Based Calendar Automation

Newer tools such as **Motion** and **Reclaim.ai** use AI and scheduling algorithms to automatically add tasks into a student's calendar. They can move study sessions when conflicts happen and help students manage their available time. However, these tools treat all calendar activities in the same way. They do not consider subject difficulty, study quality, focus levels, or student

performance. They mainly work as smart calendar tools instead of complete academic planning systems [8], [9].

2.5 Research Projects & Academic Studies

Many researchers are exploring the use of artificial intelligence and machine learning to improve study planning and personalized learning. Some systems use recommendation methods to suggest the best time for reviewing information based on student behavior and learning progress. This idea is similar to spaced repetition methods used in applications like Anki. Other research projects use large language models to create personalized study plans from course materials or generate review questions.

Research studies about AI-based learning and gamification show that modern educational systems are moving from simple fixed rules toward smarter systems that use student data, prediction methods, and personalized feedback. These approaches can improve student engagement and, in some cases, academic performance. However, many existing systems still focus on only one area: personalized learning, automatic scheduling, or motivation through gamification. They usually do not combine all these features into one connected system.

This limitation can also be seen in current educational applications. This is the main reason behind the development of **StudyNexus**, which combines AI-generated schedules, student behavior analysis, and social motivation features using one shared data system.

3. Comparison Table

Feature	Notion (+AI)	Todoist / TickTick	Forest / Study Bunny	StudySmarter	Motion / Reclaim.ai	StudyNexus
Main Purpose	Flexible workspace for notes and tasks	Task and to-do management	Focus timer with game features	Study planning and flashcards	Automatic calendar planning	AI-powered study planner

AI Schedule Creation	✗ No automatic scheduling	✓ Basic deadline support	✗ Not available	✗ Not available	✓ Strong automatic scheduling	✓ AI-generated schedules using Gemini 2.5 Flash
Focus Tracking	✗ Not available	✗ Not available	✓ Tracks study sessions	✗ Limited tracking	✗ Not available	✓ Tracks focus and study sessions
Study Performance Analysis	✗ Not available	✗ Not available	✗ Limited (session count only)	✓ Syllabus progress tracking	✗ Not available	✓ Productivity and efficiency scoring
Gamification	✗ Not available	✓ Basic points system	✓ Strong (trees, pets, rewards)	✗ Limited	✗ Not available	✓ Streaks, leaderboard, and friends
Automatic Planning	✗ Manual planning	✗ Manual task planning	✗ Not available	✗ Manual study planning	✓ Automatic calendar blocking	✓ Personalized AI planning
Platform Support	✓ Web / Desktop / Mobile	✓ Web / Desktop / Mobile	✓ Mobile only	✓ Web / Mobile	✓ Web / Desktop	✓ Flutter Mobile App + PHP Admin Portal

Table 2: Comparison of StudyNexus with Existing Study and Productivity Applications

4. How AI is Used in StudyNexus

Artificial intelligence (AI) is now used in many education and productivity applications. However, different applications use AI in different ways. Most study planners only use AI to create tasks from text or answer questions through a chatbot. In these applications, AI is only an extra feature and is not part of the whole study planning process.

Recent research shows that learning systems are moving from simple rule-based methods to AI systems that use student data, learning analysis, and personalized feedback. However, many current applications still use AI in a limited way.

StudyNexus uses AI as part of the main system instead of only as a chatbot. It uses Google Gemini 2.5 Flash for two main purposes, and both are controlled by editable prompts stored in Firestore. Administrators can update these prompts through the admin portal without changing the application code.

The first purpose is to generate a weekly study schedule based on the student's tasks, deadlines, available study time, subject difficulty, and recent study activity. The AI also creates personalized insights, such as burnout risk, focus advice, learning progress, and study consistency. The AI works with structured data from the application, and every response must follow a fixed JSON format before it is saved. This makes the results easier to check and manage.

The second purpose is rule-based analytics. StudyNexus calculates performance values such as focus score, productivity score, and efficiency score using formulas. These scores are based on study sessions, completed tasks, break time, and session performance. Combining AI with rule-based calculations makes the system more reliable because users can understand how the results are produced.

StudyNexus also includes **Nexus Analytics**, which turns study data into simple and useful information. The dashboard loads data from Firebase Firestore, processes it, and displays it in an easy-to-read interface. To improve speed, it loads study sessions, tasks, and subjects at the same time using asynchronous requests. It also checks for missing or incorrect data to prevent errors.

The dashboard shows information about study consistency, subject performance, learning progress, time management, study behavior, and subject details. It includes features such as study streaks, activity heatmaps, productivity trends, focus analysis, and schedule completion to help students understand their study habits.

The dashboard is built with Flutter's **FutureBuilder** for efficient data loading. It also checks user authentication and validates data before displaying it. A responsive heatmap and smooth scrolling allow students to view large amounts of information on different screen sizes.

By combining AI recommendations, rule-based calculations, and a fast analytics dashboard, StudyNexus helps students plan their studies, track their progress, and better understand their learning habits. This makes the system more useful, reliable, and easier to understand.

5. Technology Used

StudyNexus is developed using a combination of technologies that support a scalable, secure, and cross-platform study planning system. The application follows a two-part architecture consisting of a student mobile application and a separate administrator portal, both connected through a shared Firebase backend.

The Flutter mobile application provides the main interface for students and guest users on Android and iOS devices. It allows users to create and organize subjects, manage tasks, generate AI-based study schedules, track study sessions, and view personalized performance insights.

The PHP admin portal is a dedicated web application designed for administrators. It uses secure authentication sessions and communicates with Cloud Firestore through a protected REST API. The portal enables administrators to manage user accounts, review activity logs, and control the AI configuration, including the prompts used for generating study schedules and insights. All administrative actions are recorded through an audit log to maintain transparency and accountability.

The Firebase backend acts as the central data management layer of the system. Cloud Firestore is used to store and synchronize application data, Firebase Authentication manages user registration and secure access, and Firebase Realtime Database handles frequently changing

information such as user online status. This cloud-based structure ensures that data remains synchronized across devices while supporting real-time updates.

This architecture provides a clear separation between the student-facing application and the administrative management system. By combining a cross-platform mobile interface, a secure administration layer, and cloud-based services, StudyNexus provides a reliable environment for managing AI-powered study planning and user activity.

6. Summary of Gaps & StudyNexus's Position

The reviewed studies and applications show a clear issue: most existing tools focus on only one area, such as scheduling, focus tracking, motivation through gamification, or AI assistance. Very few tools combine all these features into one adaptive system that works across different platforms. Workspace tools like Notion provide flexibility but require more manual setup and have limited built-in analytics. Task management tools like Todoist are useful for organizing tasks but do not fully support academic needs. Gamified focus apps like Forest help students maintain attention but do not consider what subjects they are studying. AI calendar tools like Motion can automatically arrange tasks but do not consider study quality or subject difficulty. Research on AI-based learning and gamification also shows this problem and suggests that future systems should combine personalized AI support with clear feedback based on student data.

StudyNexus addresses this gap by combining AI-generated schedules and recommendations, focus and productivity measurements, streaks and leaderboards for student motivation, and a Flutter/PHP system connected through Firebase. It also adds an AI control and monitoring layer, which is not commonly found in the reviewed tools.

Chapter 3: Requirements Analysis

1. Overview

This chapter explains the requirements analysis of StudyNexus, a cross-platform academic productivity application developed using Flutter and Firebase. The system helps students manage their subjects and tasks, create study schedules, complete guided study sessions, and track their academic performance using AI-generated insights. Moreover, a web-based administration portal allows authorized administrators to manage user accounts, track system activity, and control the AI services used for study planning and personalized recommendations.

2. Requirements Gathering

The system requirements were collected through research on existing academic productivity applications and by studying common student study behaviors. This research showed several common problems, such as using different tools for different tasks, difficulty creating and following study schedules manually, limited feedback from current study timer applications, and the need for AI-based support in study planning.

The research also showed the importance of motivational features, such as tracking progress, maintaining study streaks, and using leaderboards. In addition, it showed the need for administrative features to monitor user activity and manage AI-generated information. Each requirement was checked and matched with the technologies and features used in StudyNexus, including Firebase Firestore, Firebase Authentication, Gemini AI services, and the administration dashboard.

3. User Groups

StudyNexus is designed to support two main types of users:

- Individual Students, who use the system to organize their subjects, tasks, study schedules, study sessions, and track their academic progress.

- Study Groups and Other Students, who can use features such as adding friends, sending messages, receiving notifications, and viewing leaderboards while still managing their own personal study activities.

4. Functional Requirements

The system provides the following core functional requirements:

Module	Description
Account Management	User registration, authentication, guest account support, profile management, and password recovery.
Subject Management	Creation, modification, and deletion of subjects with difficulty levels and study history tracking.
Task Management	Organization of academic tasks with priorities, deadlines, status tracking, and subject associations.
Schedule Planning	Manual scheduling or AI-generated weekly study plans based on workload, deadlines, and user availability.
Study Session Tracking	Pomodoro and manual study timers with automatic break recording and session logging.
Post-Session Reflection	Collection of user feedback regarding session difficulty, completion percentage, and overall study experience.
Performance Analytics	Calculation of focus, productivity, efficiency, consistency, and subject-specific performance metrics.
AI Study Insights	Generation of personalized recommendations and learning insights using Gemini AI.
Social Features	Friend management, messaging, and weekly study leaderboards.
Notifications	Delivery of reminder, social, achievement, system, and broadcast notifications.
Planner Export	Export of daily and weekly study schedules in PDF format.

Administration	User management, activity monitoring, suspicious activity validation, and AI configuration tools.

Table 3: StudyNexus System Modules and Their Descriptions

5. Non-Functional Requirements

The system also satisfies several important quality requirements:

- **Real-Time Synchronization:** Firestore listeners allow tasks, schedules, notifications, and study data to update instantly across the system.
- **Security:** Firebase Authentication, user roles, and CSRF protection help protect the mobile application and the admin website.
- **Reliability:** AI services use retry methods, waiting strategies, and backup solutions to keep the system working properly.
- **Performance:** Improved Firestore queries and loading data at the same time help reduce waiting time and make the system faster.
- **Usability:** A simple and consistent design, clear messages, loading signs, and empty page messages improve the user experience.
- **Scalability:** The Firebase database structure allows the system to grow and add new features in the future without major changes.
- **Auditability:** User actions and important system activities are saved using a central activity tracking system.

6. User Needs

Both user groups have distinct but overlapping needs. Individual students want one place to manage everything—subjects, tasks, schedules, study tracking, and their own performance. It's easier if all these pieces come together in a single app. They like features that lighten the load: things like AI that automatically builds a good schedule, solid productivity stats they can actually use, and maybe some motivation—streaks or achievement badges. Exporting study plans matters too, especially if they want to share with a tutor or keep a backup. Study groups and peer learners need tools that let them work together but still keep their privacy. They want to connect with friends, send messages, maybe even see a leaderboard of who's on top, or get a ping when someone unlocks a new achievement. Progress comparisons help fuel some

friendly competition, but it’s still important that no one’s grades or sensitive info gets out in the open.

7. Use Case Diagram

The use case diagram for StudyNexus breaks things down into three main roles: Student, Guest User, and Administrator. Students are the main users—once they register, they get full access to everything, from planning and tracking their studies to social features. Guest Users aren’t registered yet; they can poke around and try out basic features. If they like what they see, they can create a full account later. Administrators use a separate web portal. They handle things like managing user accounts, monitoring what’s happening on the platform, and setting up AI features.

The diagram also points out two key relationships. First, when someone uses the AI to generate a schedule, it builds on the usual manual schedule-planning—AI is like an added layer. Second, every time someone runs the study timer, there’s a built-in reflection step at the end. So, after the session wraps up, the system prompts the user to reflect before moving on.

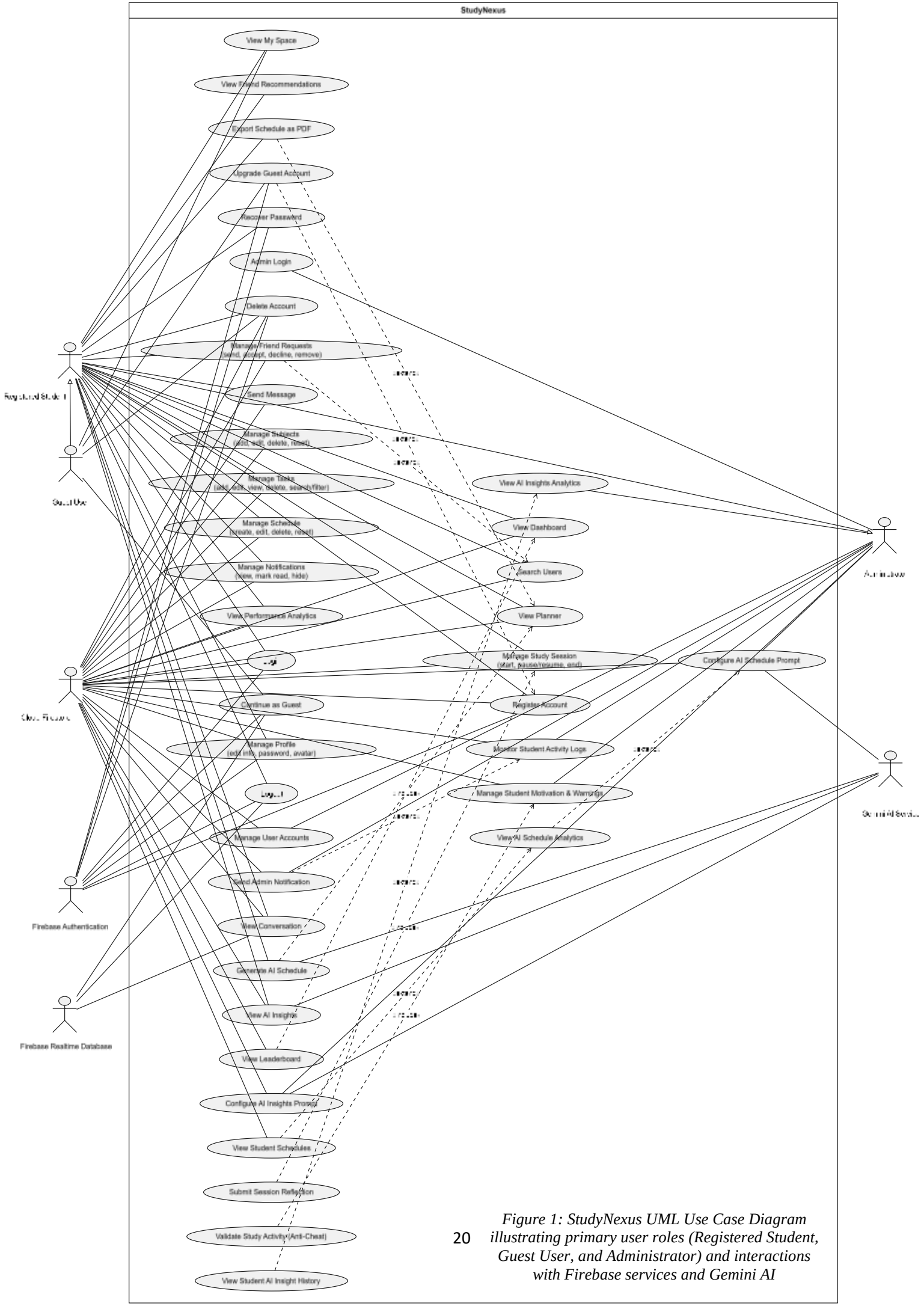


Figure 1: StudyNexus UML Use Case Diagram illustrating primary user roles (Registered Student, Guest User, and Administrator) and interactions with Firebase services and Gemini AI

8. Data Flow Diagrams (DFD Level 0 and 1)

The use case diagram of StudyNexus includes three main types of users: Students, Guest Users, and Administrators. Students are the main users of the system. After creating an account, they can manage their subjects, tasks, study schedules, and track their progress. They can also use social features like friends and leaderboards. Guest Users can try some basic features without creating an account. If they want to use more features, they can register and become students. Administrators use a separate admin portal to manage users, monitor system activities, and control AI settings.

The diagram also shows two important relationships. First, the AI schedule generation feature works together with manual scheduling. Users can create schedules by themselves or ask the AI to help them. Second, every study session includes a reflection step. After finishing a study session with the timer, the system asks users to review their session and think about their performance before continuing.

- **DFD Level 0: System Context**



Figure 2: Level 0 Data Flow Diagram (Context Diagram) of the StudyNexus System

- **DFD Level 1: Functional Decomposition**

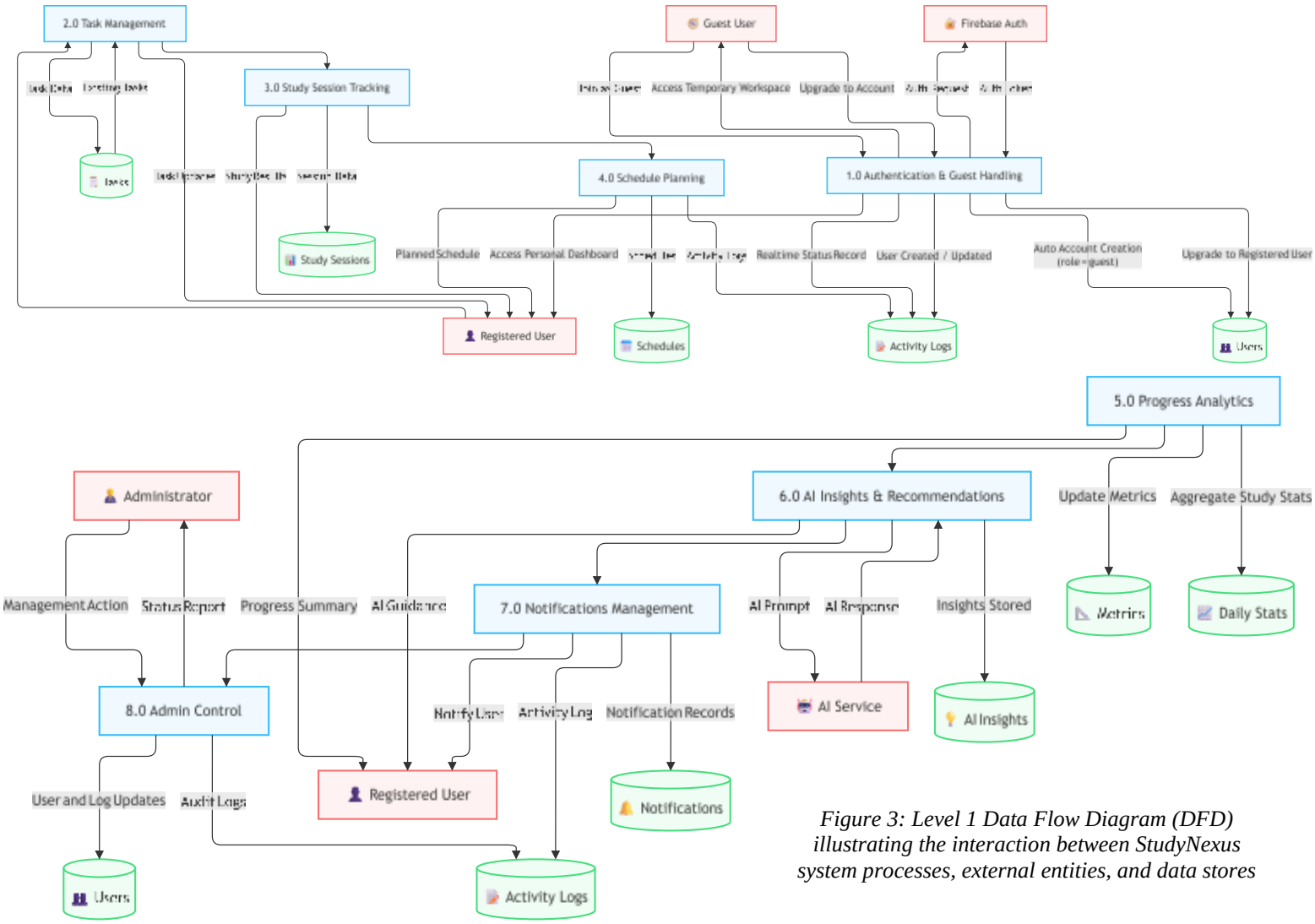
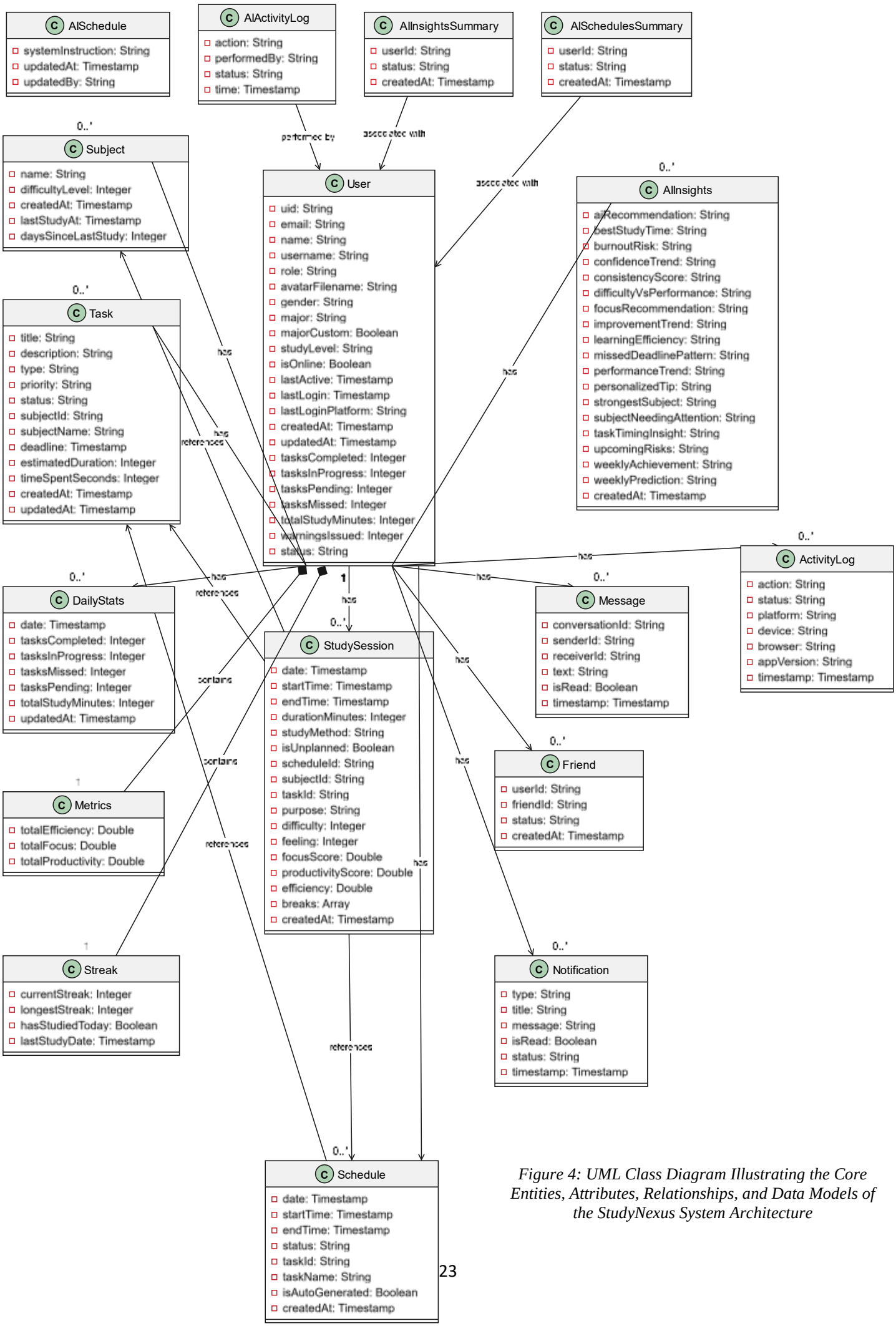


Figure 3: Level 1 Data Flow Diagram (DFD) illustrating the interaction between StudyNexus system processes, external entities, and data stores

9. Class Diagram

Here's how the core domain model for StudyNexus works: at the heart of everything is the User class. Pretty much every other piece—Subject, Task, Schedule, StudySession, DailyStat, AIInsight, Notification, Friend, Message, and ActivityLog—is tied to a specific user. So, each of these entities belongs to one user and fits inside their document in Firestore.

Tasks can link up with Subjects. As for StudySessions, they can point to a Task, a Subject, and sometimes a Schedule if they're planned ahead of time. Plus, every StudySession can include any number of StudyBreaks, so the system can keep track of each pause taken during a session.


AISchedule
systemInstruction: String
updatedAt: Timestamp
updatedBy: String
AIActivityLog
action: String
performedBy: String
status: String
time: Timestamp
AIInsightsSummary
userId: String
status: String
createdAt: Timestamp
AISchedulesSummary
userId: String
status: String
createdAt: Timestamp
Subject
name: String
difficultyLevel: Integer
createdAt: Timestamp
lastStudyAt: Timestamp
daysSinceLastStudy: Integer
User
uid: String
email: String
name: String
username: String
role: String
avatarFilename: String
gender: String
major: String
majorCustom: Boolean
studyLevel: String
isOnline: Boolean
lastActive: Timestamp
lastLogin: Timestamp
lastLoginPlatform: String
createdAt: Timestamp
updatedAt: Timestamp
tasksCompleted: Integer
tasksInProgress: Integer
tasksPending: Integer
tasksMissed: Integer
totalStudyMinutes: Integer
warningsIssued: Integer
status: String
AIInsights
aiRecommendation: String
bestStudyTime: String
burnoutRisk: String
confidenceTrend: String
consistencyScore: String
difficultyVsPerformance: String
focusRecommendation: String
improvementTrend: String
learningEfficiency: String
missedDeadlinePattern: String
performanceTrend: String
personalizedTip: String
strongestSubject: String
subjectNeedingAttention: String
taskTimingInsight: String
upcomingRisks: String
weeklyAchievement: String
weeklyPrediction: String
createdAt: Timestamp
Task
title: String
description: String
type: String
priority: String
status: String
subjectId: String
subjectName: String
deadline: Timestamp
estimatedDuration: Integer
timeSpentSeconds: Integer
createdAt: Timestamp
updatedAt: Timestamp
ActivityLog
action: String
status: String
platform: String
device: String
browser: String
appVersion: String
timestamp: Timestamp
DailyStats
date: Timestamp
tasksCompleted: Integer
tasksInProgress: Integer
tasksMissed: Integer
tasksPending: Integer
totalStudyMinutes: Integer
updatedAt: Timestamp
Message
conversationId: String
senderId: String
receiverId: String
text: String
isRead: Boolean
timestamp: Timestamp
StudySession
date: Timestamp
startTime: Timestamp
endTime: Timestamp
durationMinutes: Integer
studyMethod: String
isUnplanned: Boolean
scheduleId: String
subjectId: String
taskId: String
purpose: String
difficulty: Integer
feeling: Integer
focusScore: Double
productivityScore: Double
efficiency: Double
breaks: Array
createdAt: Timestamp
Friend
userId: String
friendId: String
status: String
createdAt: Timestamp
Metrics
totalEfficiency: Double
totalFocus: Double
totalProductivity: Double
Streak
currentStreak: Integer
longestStreak: Integer
hasStudiedToday: Boolean
lastStudyDate: Timestamp
Notification
type: String
title: String
message: String
isRead: Boolean
status: String
timestamp: Timestamp
Schedule
date: Timestamp
startTime: Timestamp
endTime: Timestamp
status: String
taskId: String
taskName: String
isAutoGenerated: Boolean
createdAt: Timestamp
performed by
associated with
associated with
has
has
has

10. Summary

This chapter takes a close look at what StudyNexus really needs. First, I looked into common problems students face when planning their studies and broke down what both individual students and groups actually want from a tool like this. Then I set out the main features: account management, tracking subjects and tasks, planning schedules, getting help from AI for schedules and coaching, running study sessions, analytics for student performance, plus admin controls. On top of these, there are things like real-time updates, security, reliability, good performance, ease of use, the ability to scale up, and tracking changes behind the scenes.

The Use Case Diagram spells out how students, guests, and admins actually use the system, and the Class Diagram turns the Firestore data setup into a clear map of the main pieces and how they fit together. Together, these diagrams form the backbone of the system—the parts that everything else in the upcoming chapters is built on.

Chapter 4: System Design

1. Introduction

Let's dig into how StudyNexus is actually built. The idea here is pretty simple: take all those requirements and turn them into a clear, solid technical plan. StudyNexus brings together a few key pieces—a cross-platform app for students built with Flutter, a PHP dashboard for admins, and Firebase for everything from logging in to storing data. On top of that, Gemini AI handles smart scheduling and academic insights. This chapter lays out how everything fits together: how the system is structured, how the database works, and how all these moving parts connect. The goal? Make sure the platform can grow, stay tidy, and let the different sections communicate smoothly.

2. Database Design

Under the hood, StudyNexus relies on Google Cloud Firestore as its main database. Firestore works a bit differently from traditional databases—it's NoSQL and organizes information as collections (think folders) filled with individual documents (like files), which hold all the details about users and app data.

The main collections include:

Collection	Purpose
Users	Stores user profiles, roles, and productivity statistics
Subjects	Stores academic subjects and study information
Tasks	Stores assignments, exams, and study tasks
Schedules	Stores planned study sessions generated manually or by AI
Study Sessions	Stores completed study activities and performance metrics
AI Insights	Stores AI-generated recommendations and analysis
Notifications	Stores system and user notifications
Friends/Messages	Supports social interaction features

Daily Stats	Stores aggregated daily productivity data
Activity Logs	Tracks user activity and security information

Table 4: High-Level Firestore Database Schema of the StudyNexus Application, Representing the Main Collections and Their Functional Responsibilities

3. Data Relationships and Access Patterns

Although Firestore is a NoSQL database, relationships between entities are maintained through document references and identifiers.

Examples:

- A user can have multiple subjects, tasks, schedules, and study sessions.
- Tasks can be linked to specific subjects.
- Study sessions reference completed tasks and subjects.
- AI insights are generated from collected user activity data.

Sub-collections are used for frequently accessed user-specific data, improving organization and query efficiency.

4. System Architecture

The StudyNexus architecture consists of four main layers:

Client Layer

The client layer contains:

Flutter Mobile Application

- Student interface
- Task management
- Study timers
- Planner
- Performance analytics
- AI features

PHP Administrative Dashboard

- User management
- System monitoring
- Activity supervision
- AI configuration management

Backend and Cloud Layer

Firebase provides:

- Authentication
- Cloud Firestore database
- Real-time data synchronization
- Secure user data management

The administrative dashboard communicates with Firestore through server-side authentication.

AI Layer

Google Gemini AI provides intelligent features including:

- Automatic study schedule generation
- Academic performance analysis
- Personalized recommendations

AI services analyze student activity data and return structured insights stored in Firestore.

5. Summary

The system architecture is divided into four main parts: the student application, the administration system, the database, and the AI services. This separation makes the system easier to manage, more secure, and easier to improve in the future. It also allows adding new features later, such as advanced reports and more AI functions.

Chapter 5: System Implementation

1. Introduction

This chapter presents the implementation of the StudyNexus platform, a cross-platform academic productivity system composed of a Flutter-based mobile application, a Firebase cloud backend, a PHP-based web administration panel, and an artificial-intelligence layer built on Google's Gemini large language model. It documents the development environment and technology stack, the structure of the Cloud Firestore database, and the implementation of the system's core modules — authentication, subject and task management, study session tracking, AI-driven scheduling and analytics, social features, real-time presence tracking, and the administrative dashboards used to supervise and moderate the platform.

2. Development Environment and Technology Stack

StudyNexus pulls together a mobile app, a cloud backend, an admin website, and a generative AI service into one unified client-server system. Both the mobile app and the admin panel share a single Cloud Firestore database.

On the user side, everything runs on Flutter and Dart. There's just one codebase, compiled natively for each platform. It handles all views—Home, Friends, Messages, Subjects, Tasks, Planner, Study Session, AI, Performance Analytics, and My Space. The app relies on Flutter’s stateful widget lifecycle (initState, setState, mounted) to load Firebase data asynchronously and keep the UI fresh in real time.

For the backend, Firebase does the heavy lifting. Authentication covers both email/password logins and anonymous guest sessions you can later upgrade. Cloud Firestore serves as the main NoSQL database, while Realtime Database handles quicker, lighter duties like numbering guest accounts and tracking who’s online. The Flutter app talks to Firebase through official SDKs, while the PHP admin panel connects straight to Firestore through its REST API.

The admin panel itself runs on PHP and JavaScript. Pages are server-rendered with access control, and the panel talks to Firestore as a trusted service account (since there’s no Firebase

SDK for PHP). On the frontend, JavaScript handles searching, filtering, sorting, and dynamic forms, all without reloading the page.

Generative AI comes from Google Gemini 2.5 Flash, which builds weekly study schedules and delivers customized coaching. It always sticks to strict JSON output—validated every time before getting saved. Admins can tweak the AI's behavior just by editing system prompts stored in Firestore config documents, no redeploy needed.

For extra tools: PHP cURL manages REST calls, Flutter adds PDF export through its pdf/printing packages, Firebase SDKs run throughout, and Google OAuth 2.0 authenticates service accounts.

3. Database Implementation

StudyNexus is built on Cloud Firestore, a NoSQL document database that holds all the app's data. At the heart of everything is the users collection—each user gets a unique Firebase Authentication UID. Inside each user, you'll find sub-collections like dailyStats, activityLogs, subjects, tasks, notifications, and aiInsights. There are also a few shared top-level collections, where each document points back to the user who owns it.

This setup makes things fast. The app grabs all the info it needs about a user with just a few queries. And since Firestore updates in real time, any changes show up instantly in the app's interface. No waiting around—everything just stays in sync.

Collection	Description
users	Profile data (name, username, role, status, avatar), denormalized productivity counters (tasksCompleted, tasksPending, totalStudyMinutes), and nested metrics/streak maps.
users/{uid}/dailyStats	Per-day aggregated statistics (tasks completed, missed, pending, in progress; total study minutes), keyed by date.
users/{uid}/activityLogs	Individual user activity events (login, logout, feature usage) with device, platform, and app-version metadata.
users/{uid}/notifications	Personalized reminders, social alerts, congratulatory, system, and broadcast messages.

users/{uid}/aiInsights	AI-generated coaching insights from the Gemini-based insight service, one document per day.
subjects	Academic subjects with difficulty level and study-recency indicators.
tasks	Homework, assignments, projects, and exams, with priority, status, deadline, and time spent.
schedules	Planned or AI-generated study blocks with start/end time, linked task, and completion status.
studySessions	Completed sessions with duration, timer method, recorded breaks, and focus/productivity/efficiency scores.
friends / messages	Friendship relationships (pending, accepted, blocked, removed) and direct messages between friends.
config	Global settings, including the Gemini system instructions (gemini, gemini2) used for AI scheduling and insight generation.
aiInsightsSummary / aiSchedulesSummary / ai_activity_log	Audit collections recording AI insight generation outcomes, AI schedule generation outcomes, and administrator configuration changes.

Table 5: Overview of the Firestore Database Collections, Data Organization, and Purpose within the StudyNexus System

4. User Account Management

The user account system manages guest, student, and administrator access. Firebase Authentication is used for login and password management, while Firestore stores user profiles and roles.

When the app starts, it checks the current session and directs the user to the correct page. Users can log in, create accounts, or recover passwords. The system validates user information, prevents invalid access, updates login records, and saves account activity. Registration also creates user profiles, generates unique friend codes, and supports converting guest accounts into registered accounts.

5. Mobile Application Implementation

Once authenticated, users reach the main dashboard (user_page.dart), which swaps between ten feature views via a bottom navigation bar and setState.

Index	View	Purpose
0	HomeView	Overview, live statistics, background data maintenance
1	FriendsView	Friend management and recommendations
2	MessagesView	Direct messaging between friends
3	SubjectView	Subject organization
4	TasksView	Task management
5	PlannerView	Study planning, incl. AI-generated schedules
6	StudySessionView	Session configuration and timer tracking
7	AIView	AI-powered personalized study insights
8	AnalyticsView	Productivity and performance analytics
9	MyspaceView	Profile and account settings

Table 6: Description of StudyNexus Application Views and Their Functional Purposes

Home, Profile, and Presence

HomeView is the default screen of the app. It keeps user data updated in real time by monitoring profiles, daily statistics, and notifications. It also performs background tasks such as updating study progress, streaks, overdue tasks, schedules, subject activity, and leaderboard rankings.

MyspaceView allows users to manage their profiles. Guests can create an account, while registered users can edit their information, change sensitive details securely, or delete their account. The app also provides unlockable avatars based on study progress.

A presence tracker monitors user activity by detecting app status changes and sending regular updates to Firebase. The admin panel uses the same data to display user online status and activity.

Subject and Task Management

SubjectView shows the user's subjects in real time and allows adding, editing, renaming, or deleting subjects. When a subject is renamed or removed, the related tasks are updated automatically using batch operations. TasksView helps users manage tasks by tracking different states such as pending, in progress, completed, overdue, and missed. It also provides search and filtering options. The "Start All Over" feature can clear all study data, including subjects, tasks, and schedules.

Study Session Management

Users can create study sessions either from a planned schedule or by manually choosing a subject and task. TimerView manages the study timer using either a Pomodoro mode with automatic breaks or a manual timer. After each session, the system saves important details such as study duration, focus level, productivity, and efficiency scores.

Planner, AI Scheduling, and PDF Export

PlannerView allows students to manage their study plans using Day, Week, and Month calendar views. Users can manually add schedules, and the system checks important details such as missing information, incorrect times, and schedule conflicts before saving. The AI scheduling feature uses student data, including tasks, exams, study history, and available free time, to create a balanced study plan using Gemini 2.5 Flash. The AI follows rules such as prioritizing deadlines, avoiding study during sleeping hours, keeping suitable session durations, and applying spaced repetition. If the AI request fails, the system retries automatically. The generated schedule is then reviewed and saved. The app also supports PDF export, allowing users to share daily or weekly study plans with clear formatting, status labels, and AI-generated schedule indicators.

Social Features and Notifications

StudyNexus includes social tools that help students connect and stay motivated. The Friends feature allows users to send, accept, block, or remove friends. It also suggests new friends

based on shared connections, similar study information, and recent activity. The messaging feature allows students to chat privately using real-time updates. The leaderboard compares users and friends based on weekly study time, showing top performers and notifying users when their rank changes. The notifications system organizes reminders, social updates, achievements, and system messages, allowing users to view, read, or hide notifications easily.

6. Artificial Intelligence and Analytics Implementation

The AI feature provides students with personalized study recommendations generated by Gemini AI. The system uses a cache system to avoid creating new insights too often. It collects recent data about study sessions, tasks, and subjects, then analyzes patterns such as consistency, performance changes, study habits, and burnout risk. The generated results are checked, cleaned, and stored in the database for future use.

The Performance Analytics dashboard collects data from sessions, tasks, and subjects using Firestore. It safely processes the data by handling missing or incorrect values to prevent errors. The system then calculates useful statistics and displays them through visual elements such as consistency tracking and performance comparisons, helping students understand their study progress.

7. Web-Based Administration Panel Implementation

The PHP administration panel uses the same Firestore database as the mobile application but connects through secure REST API calls. Each admin page is protected by AdminGuard.php, which checks the administrator login, manages session timeout after inactivity, and prevents cached pages from being reused. Important actions use security measures such as CSRF tokens to protect against unauthorized requests, and updates follow the Post/Redirect/Get method to avoid repeated submissions.

GoogleAuth.php handles Firebase authentication by creating secure access tokens and allowing the system to communicate with Firestore. It provides reusable functions for reading, searching, adding, and updating database data through REST requests.

The **administration panel** contains several main sections. The **Manage Accounts page** allows administrators to view, search, filter, and enable or disable user accounts while preventing them from disabling their own account. The **Study Data Monitoring page** allows administrators to

check user activity records and send direct notifications. The **Admin Motivation dashboard** shows study activity rankings, detects suspicious study data, and allows administrators to send warnings, suspend accounts, or send motivational messages. The **AI Supervisor dashboards** allow administrators to manage AI instructions, review AI usage and results, and record all configuration changes for tracking purposes.

This system provides secure management tools while keeping the mobile application and administration panel connected through the same database.

8. Data Validation and Security

Security is implemented in several layers. First, user input is checked before any data is saved to make sure it has the correct format, length, required information, and no duplicates. Firestore uses controlled updates and batch or transaction operations to keep related data consistent. AI responses are also checked, and only approved fields are accepted while invalid data is removed. Important actions by users and administrators are recorded for auditing and debugging. Authentication errors use general messages to avoid revealing account information. In addition, sensitive actions, such as changing important account details or deleting an account, require the user to authenticate again.

9. Summary

This chapter explained how StudyNexus was implemented using its main technologies and system modules. The Flutter/Dart mobile application provides features such as user authentication, subject and task management, planning, AI-based scheduling, study session tracking, social features, notifications, and performance analysis. The application uses Cloud Firestore to store and manage the system data.

The AI features use Gemini 2.5 Flash to create personalized weekly study schedules and provide academic insights. These features use stored system instructions, structured responses, validation, and retry mechanisms to improve reliability. The PHP administration panel connects to Firestore to manage users, monitor activities, enforce anti-cheating measures, and manage AI settings. Security measures such as CSRF protection, session controls, and activity logging are also implemented.

Chapter 6: Testing and Results

1. Website Testing



Figure 5: Landing page of the StudyNexus website, introducing the platform's AI-powered study intelligence

Home

Here's the home base for StudyNexus admins—dark theme, a dash of purple, and everything built to help you manage things without any guesswork.

The regular StudyNexus dashboard aims at students—planning, AI-driven advice, connecting with friends. But the admin side? Whole different game. Here, you're in charge of supervising the system. You get tools to manage users, sift through data, and keep an eye on how everything's running.

Navigation's straightforward: plenty of white space, sharp text, and just enough color to keep things lively but professional. All the clean, modern feel of StudyNexus stays intact—just tweaked for a control room vibe.

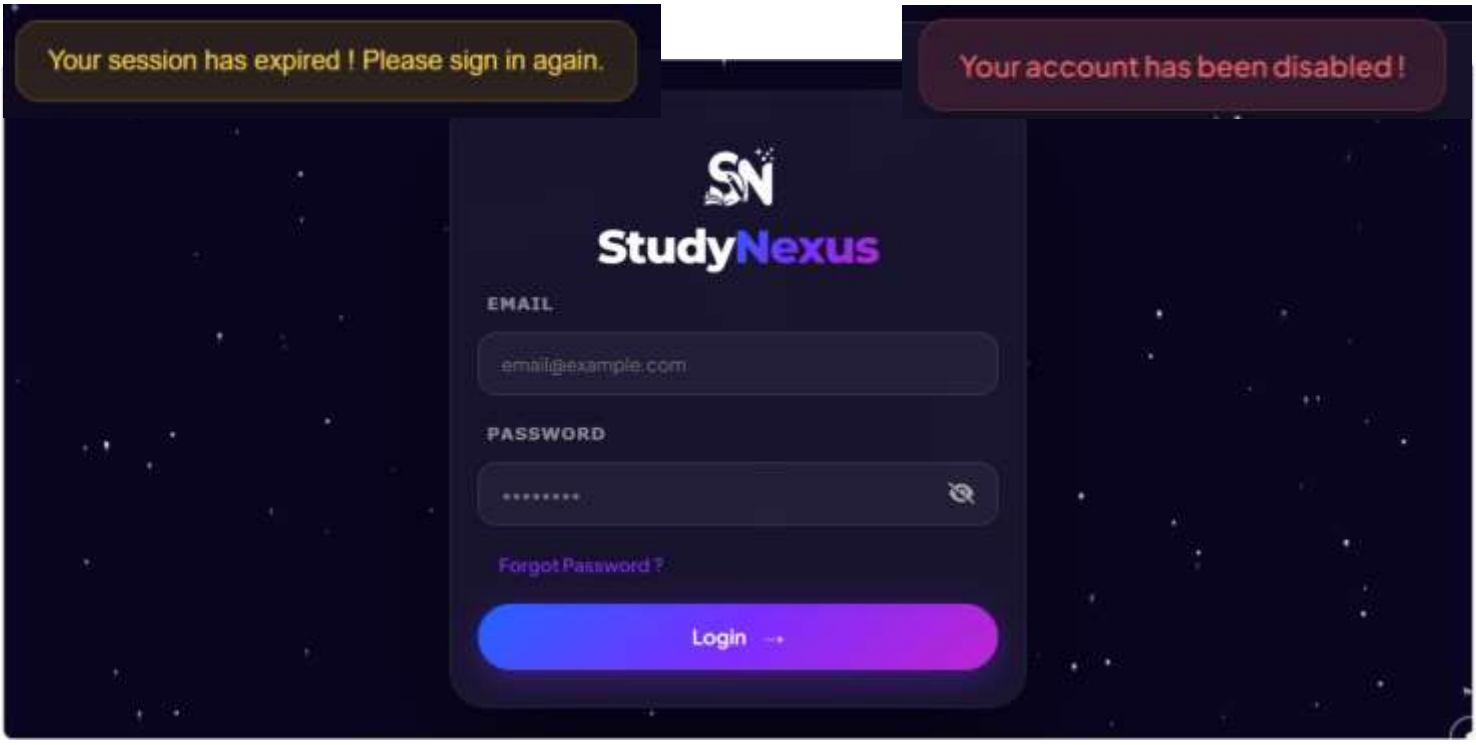


Figure 7: StudyNexus Admin Login Interface for administrator authentication with toast messages

Login: Authenticates administrators through Firebase Authentication by validating credentials, handling incorrect login attempts, and verifying account authorization. Disabled administrator accounts are blocked from accessing the system. Successful logins redirect authorized administrators to the Admin Dashboard, while inactivity monitoring automatically expires sessions after 30 minutes and returns users to the login page.



Figure 6: StudyNexus password recovery interface with success and error toast notifications

Password Reset: Allows admins to securely request a password reset, sending a Firebase Authentication reset link to their registered email address for account recovery.

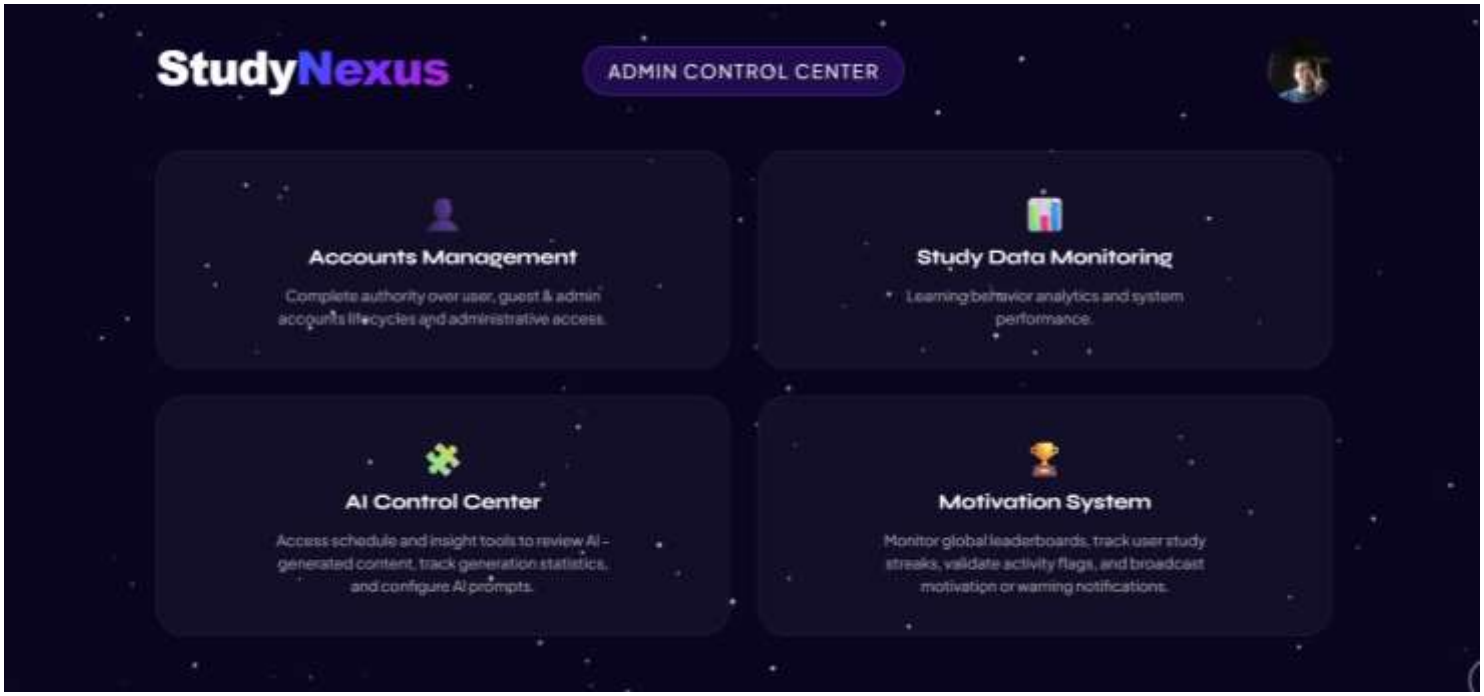


Figure 8: StudyNexus Admin Dashboard showing the main administrative modules and system management features

Admin Control Center Dashboard

gives administrators one place to manage and monitor the system. From this dashboard, administrators can manage user accounts, check study activity, control AI features, and monitor user engagement. The admin profile icon is shown as a circular avatar in the top-right corner of the header and provides access to profile options.

The main features are:

- Account Management: Manage user, guest, and admin accounts, including their access and status.
- Study Data Monitoring: View users' study activities, performance, and system activity.
- AI Control Center: Review AI-generated schedules and insights and manage AI settings.
- Motivation System: Monitor leaderboards, study streaks, user activity, and notifications.



Figure 9: StudyNexus user profile management interface displaying account details, avatar settings, and profile options

Administrator Profile Management

Displays key account details such as full name, email address, gender, and creation date. It also provides profile management options through **Edit Profile**, **Change Avatar**, and **Logout** buttons.

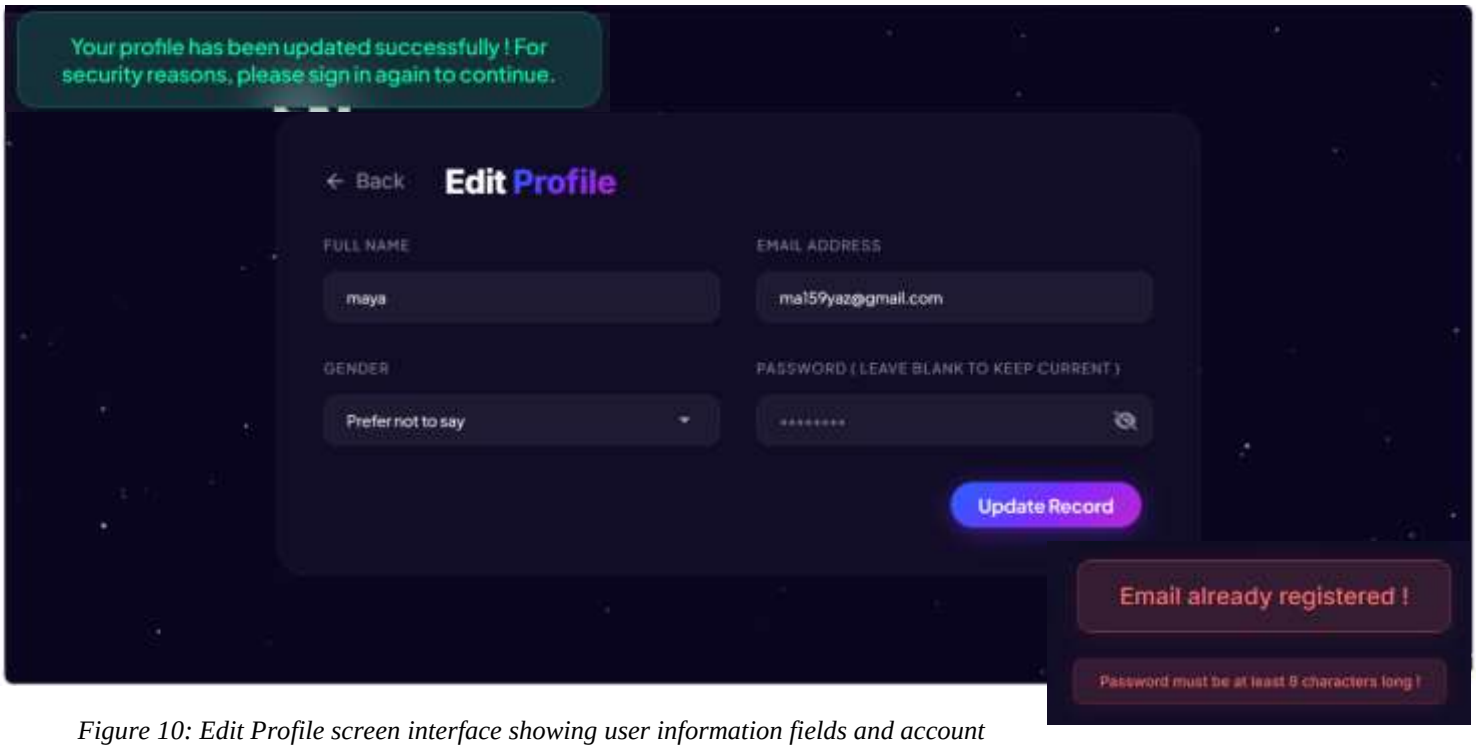
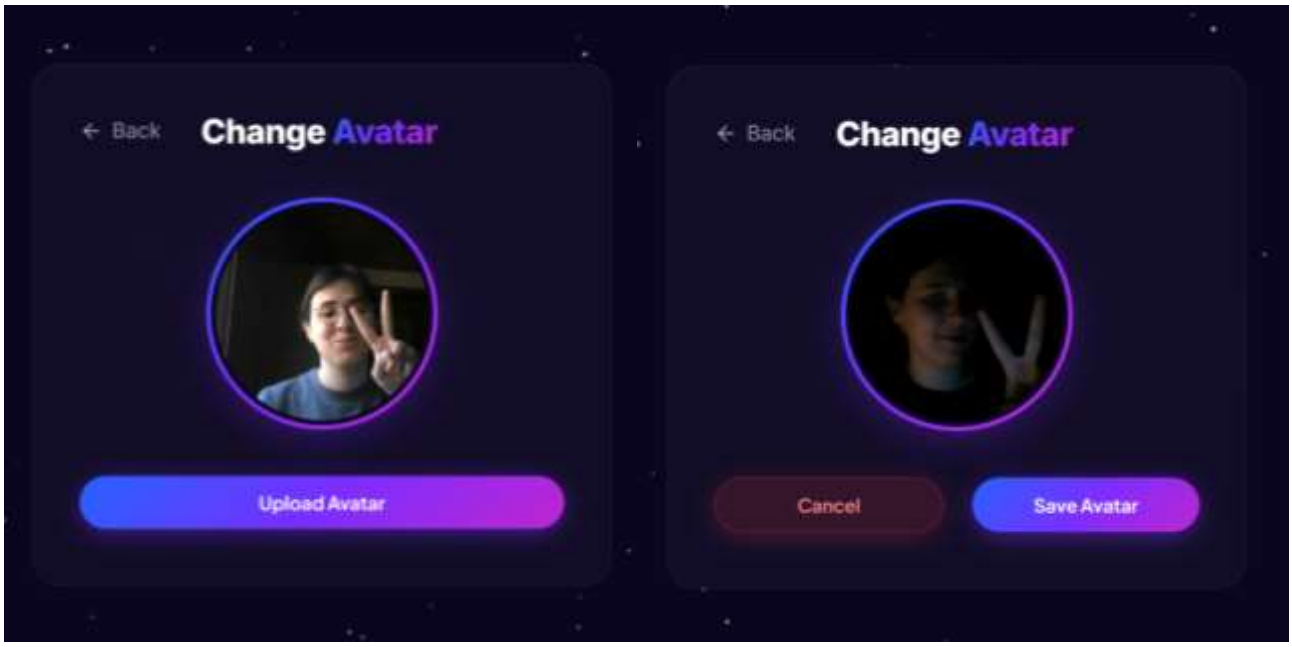


Figure 10: Edit Profile screen interface showing user information fields and account update options

After updating the profile information, the system validates the entered data, including email uniqueness and password requirements. Upon successful validation, the administrator session is securely terminated, the user is logged out, and they are redirected to the login page to authenticate again.



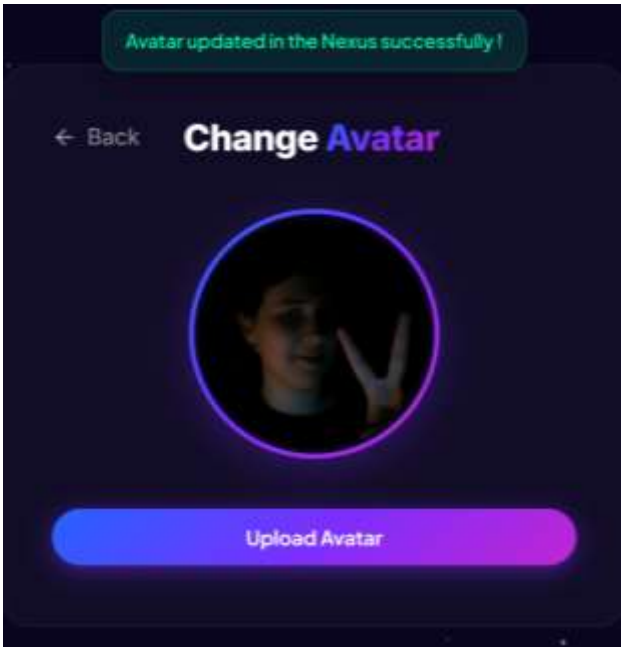


Figure 11: StudyNexus Change Avatar interface flow, showing avatar upload, preview, and success/error states

Upload Avatar Page Description:

The Upload Avatar page allows administrators to manage their profile image by displaying a preview of the current avatar and providing the option to select a new image. When a new

avatar is chosen, the system updates the preview temporarily and prompts the administrator to either cancel the change or save it. The avatar path in Firestore is only updated after confirmation, ensuring that unintended changes are avoided. Each administrator account is assigned a single avatar file linked to the account UID, allowing the new image to overwrite the existing avatar instead of creating unnecessary image history.

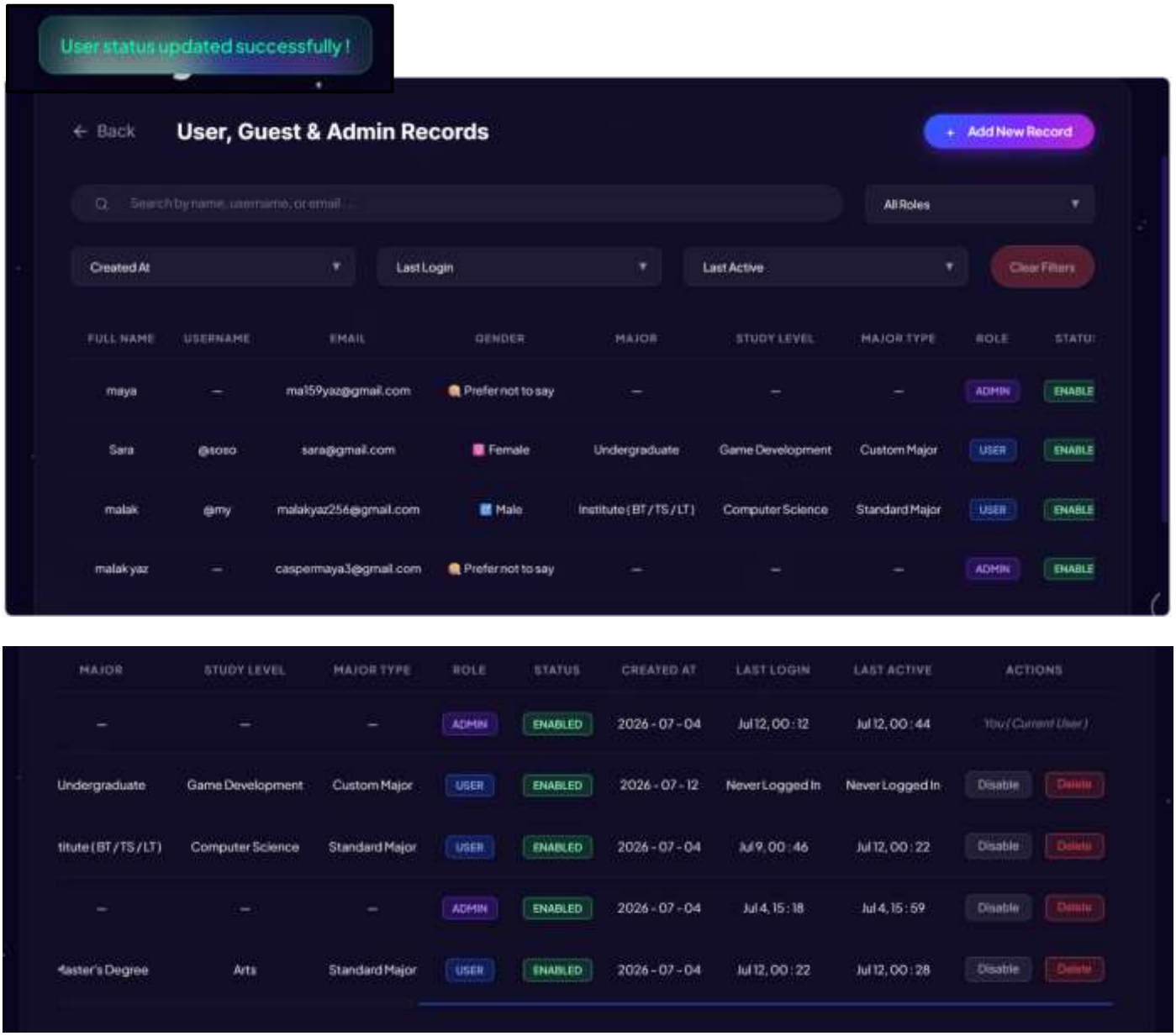


Figure 12: User, Guest, and Admin Records dashboard interface, featuring search, filtering options, user information table, and management actions

Manage Accounts Overview:

The Manage Accounts page allows administrators to efficiently manage system accounts by viewing account details, creating new user or administrator accounts, enabling or disabling

accounts, and deleting accounts. The interface ensures consistent profile data by using predefined major selections, while allowing custom major entry when the required option is unavailable. Administrator accounts are handled separately from user accounts and contain only administrative information and permissions without academic or performance-related data. Missing activity records are displayed as **"Never Logged In"** to indicate that the account has been created successfully but has not yet been accessed by the user. For security, administrators cannot disable or delete their own active account, as these actions are hidden for the currently authenticated administrator.

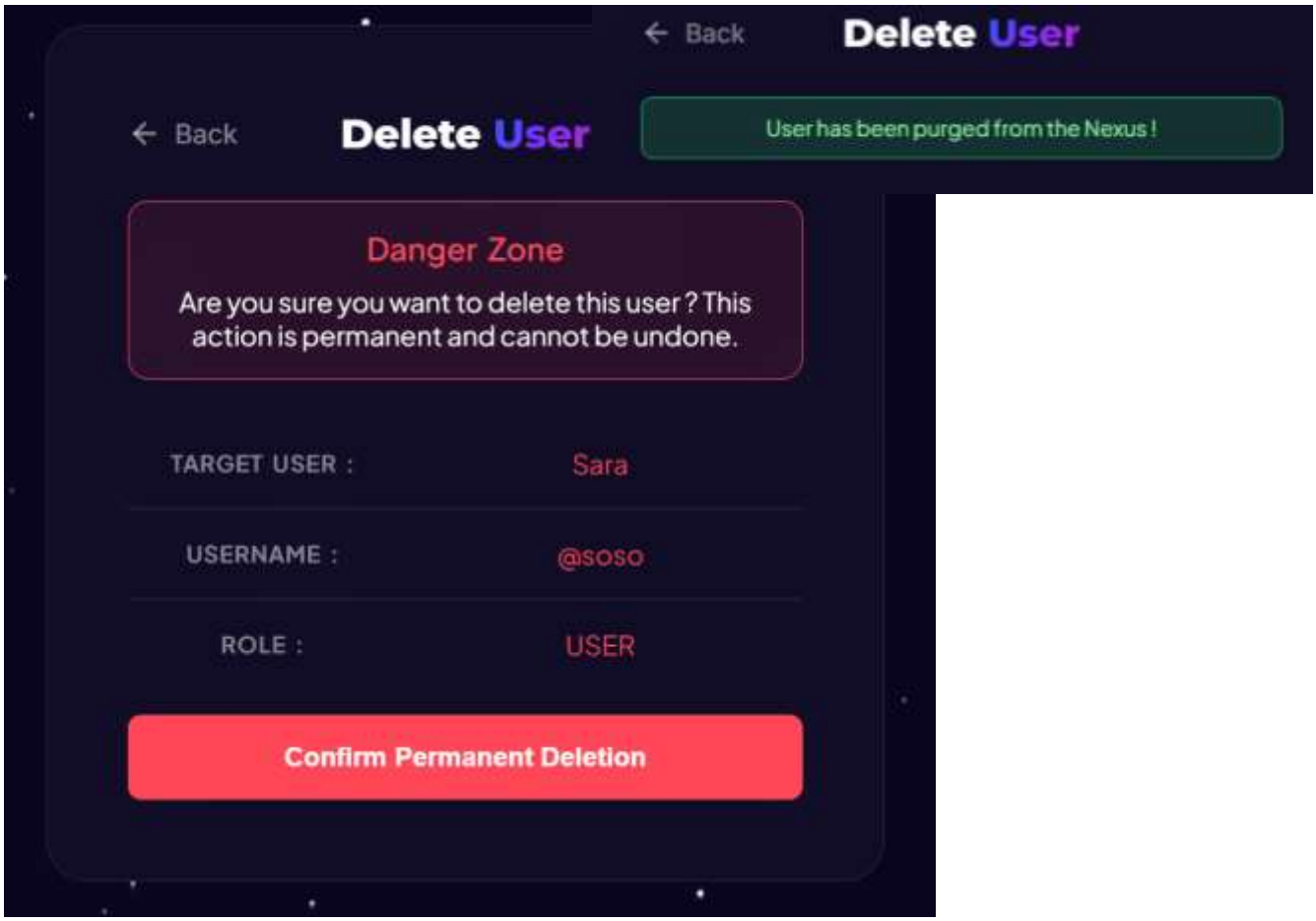


Figure 13: Delete User confirmation interface with user details and permanent deletion action

Delete Account Confirmation:

The Delete Account Confirmation page provides administrators with a verification step before permanently removing an account. It displays the target user's key information, including the username and account role, allowing administrators to confirm that the correct account is selected before proceeding with the deletion.

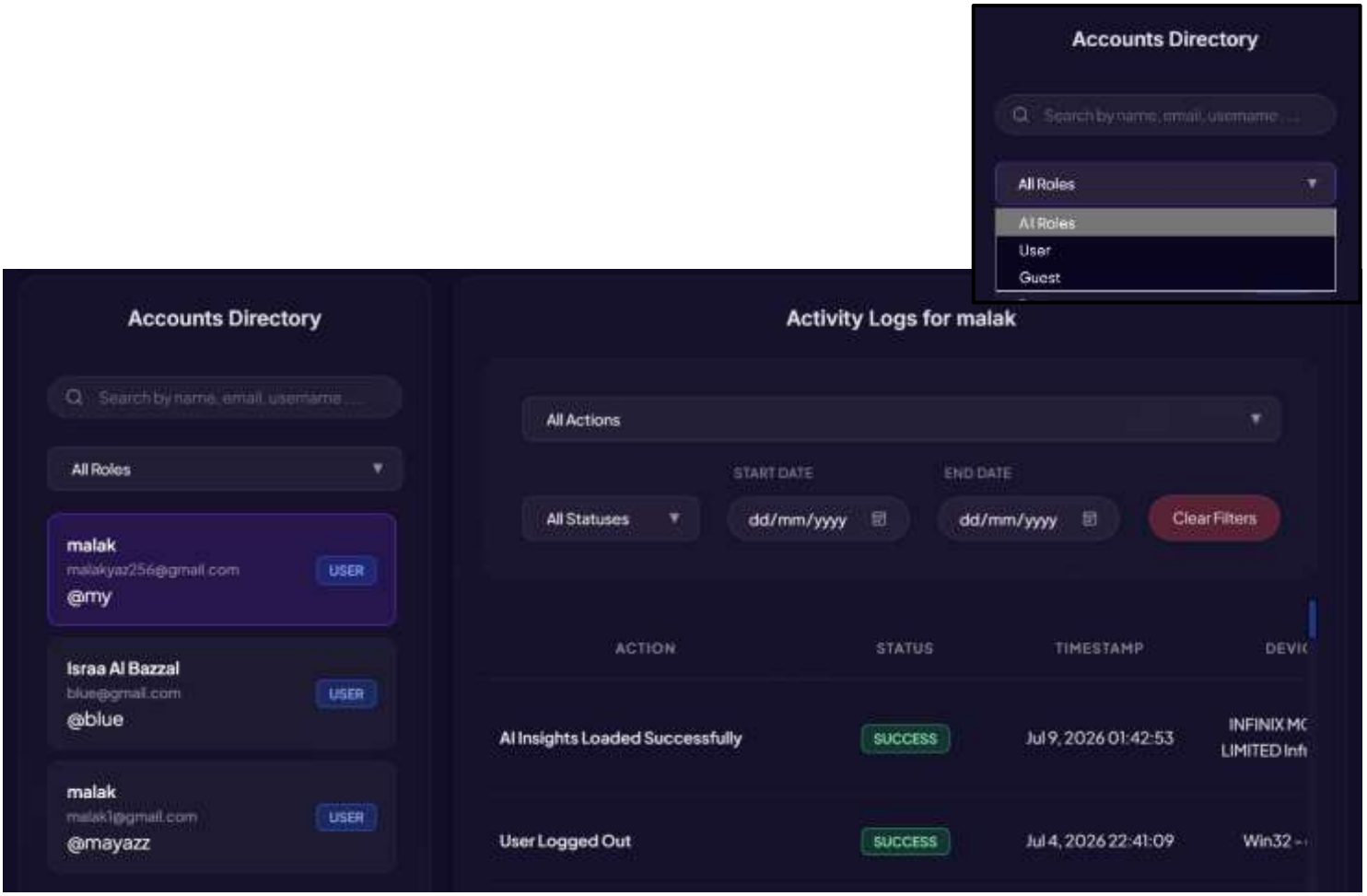


Figure 14: Activity Logs interface showing user management, activity tracking, and failure message notifications

Study Data Monitoring Dashboard

This dashboard provides administrators with comprehensive monitoring and management capabilities for registered users and guests within the StudyNexus system. It allows administrators to retrieve and view user records, perform real-time searches by name, username, or email, and filter accounts based on user roles, excluding administrator accounts that do not contain activity logs. The dashboard enables viewing selected users' or guests' activity history with dynamic filtering options based on action type, status, and date range. Filter options are automatically generated from available activity log data to support efficient analysis and tracking.

Additionally, the dashboard includes an administrative messaging modal that displays relevant user context and allows administrators to quickly send support notifications directly to a user's or guest's notification collection when a failed system action or issue requires attention.

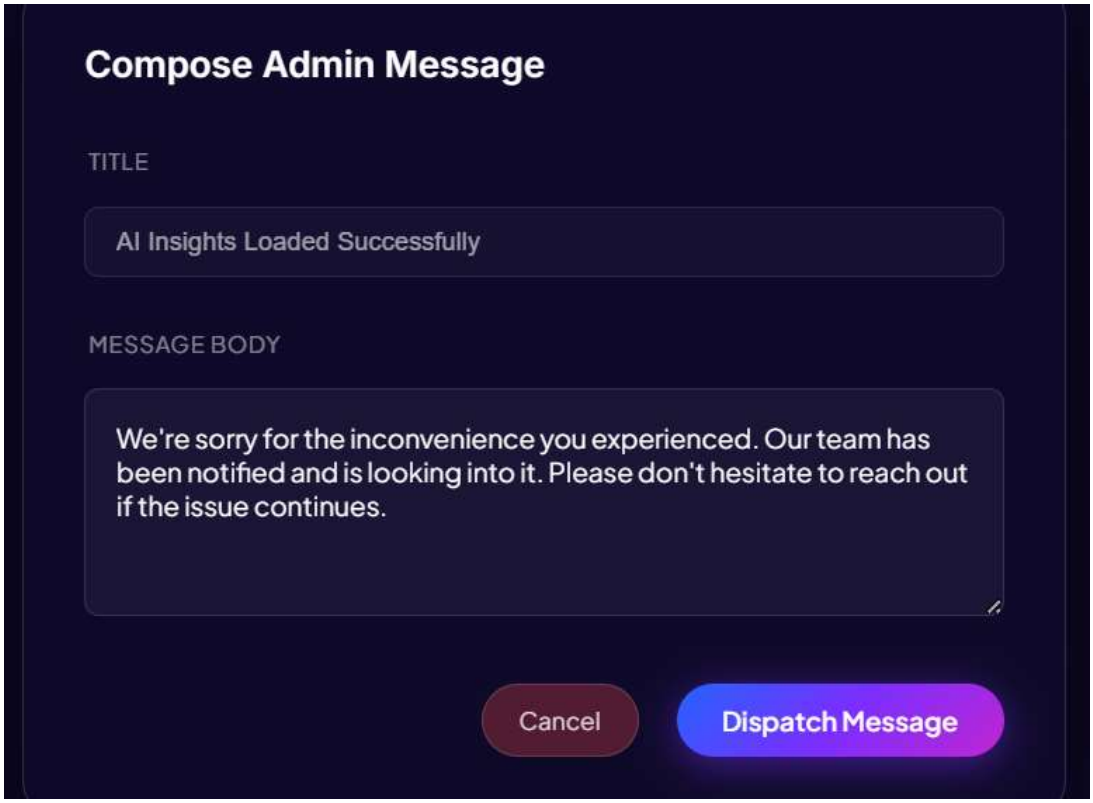


Figure 15: Compose Admin Message interface for creating and sending failure notifications to users

AI Settings Page Description:

This page provides administrators with a centralized interface to monitor and manage the AI-driven features of the StudyNexus system. It includes dedicated navigation buttons for accessing the AI Insights and AI Schedules pages, allowing administrators to review AI-generated performance analysis, monitor intelligent scheduling activities, and oversee the effectiveness of AI-assisted learning recommendations.

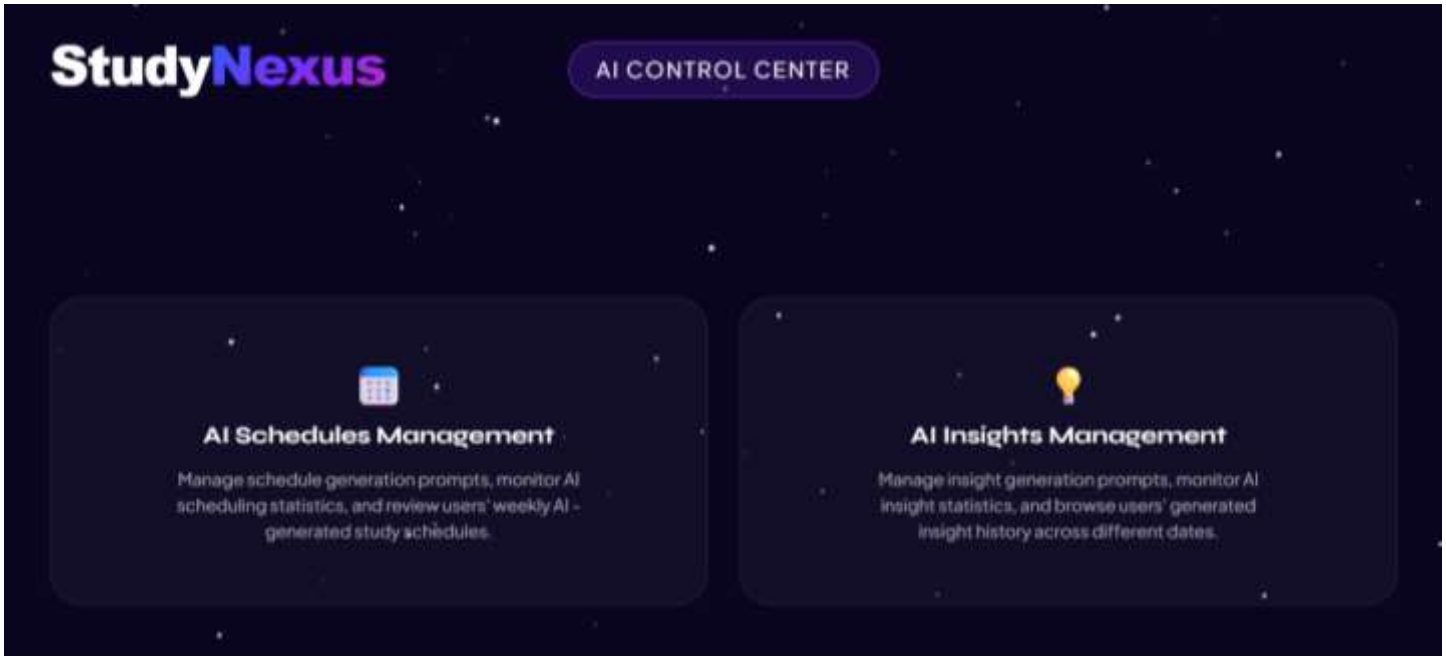
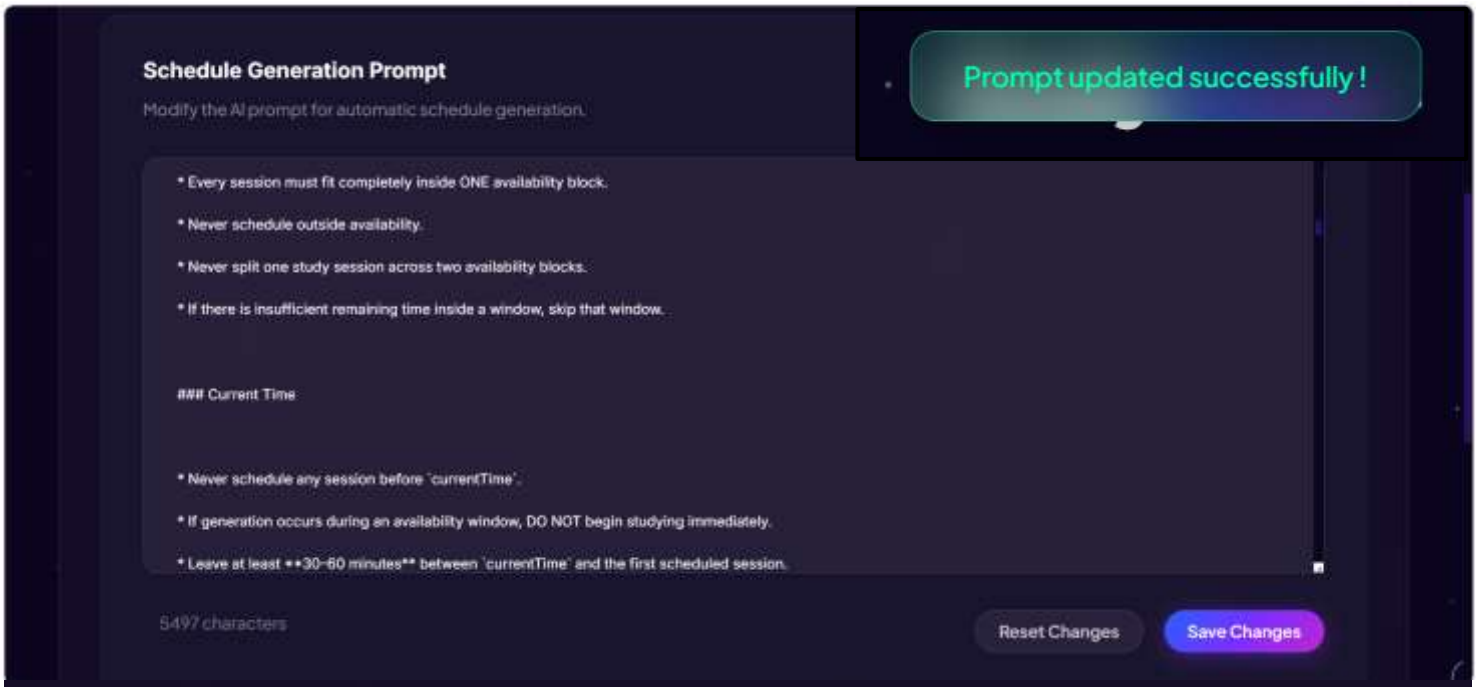
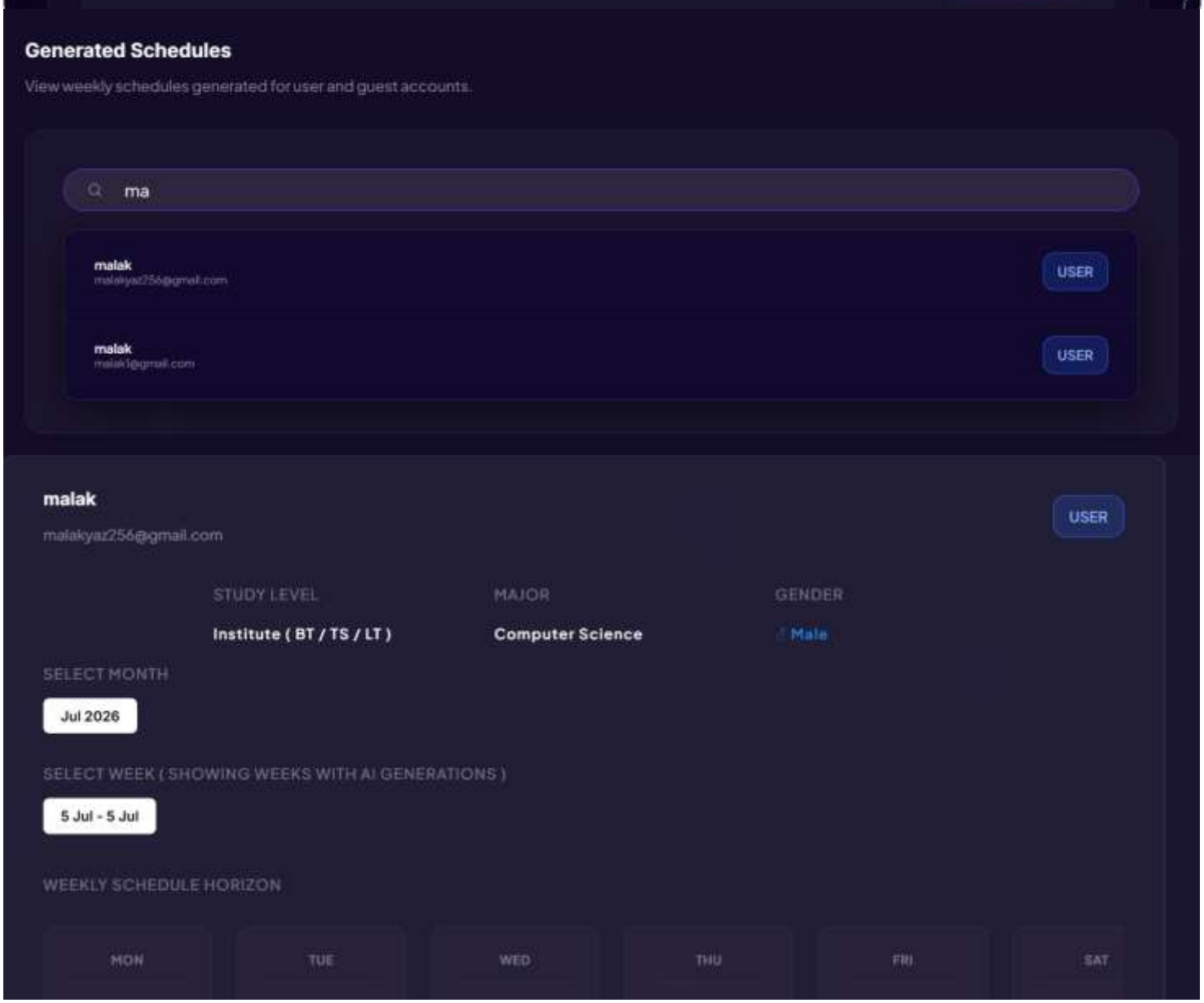


Figure 16: StudyNexus AI Control Center interface with AI schedule and insight management panels


Schedule Generation Prompt
Modify the AI prompt for automatic schedule generation.
Prompt updated successfully !
* Every session must fit completely inside ONE availability block.
* Never schedule outside availability.
* Never split one study session across two availability blocks.
* If there is insufficient remaining time inside a window, skip that window.
Current Time
* Never schedule any session before `currentTime`.
* If generation occurs during an availability window, DO NOT begin studying immediately.
* Leave at least **30–60 minutes** between `currentTime` and the first scheduled session.
5497 characters
Reset Changes
Save Changes


Generated Schedules
View weekly schedules generated for user and guest accounts.
ma
malak
USER
malak
USER
malak
malakyaz256@gmail.com
USER
STUDY LEVEL
Institute (BT / TS / LT)
MAJOR
Computer Science
GENDER
Male
SELECT MONTH
Jul 2026
SELECT WEEK (SHOWING WEEKS WITH AI GENERATIONS)
5 Jul - 5 Jul
WEEKLY SCHEDULE HORIZON
MON
TUE
WED
THU
FRI
SAT

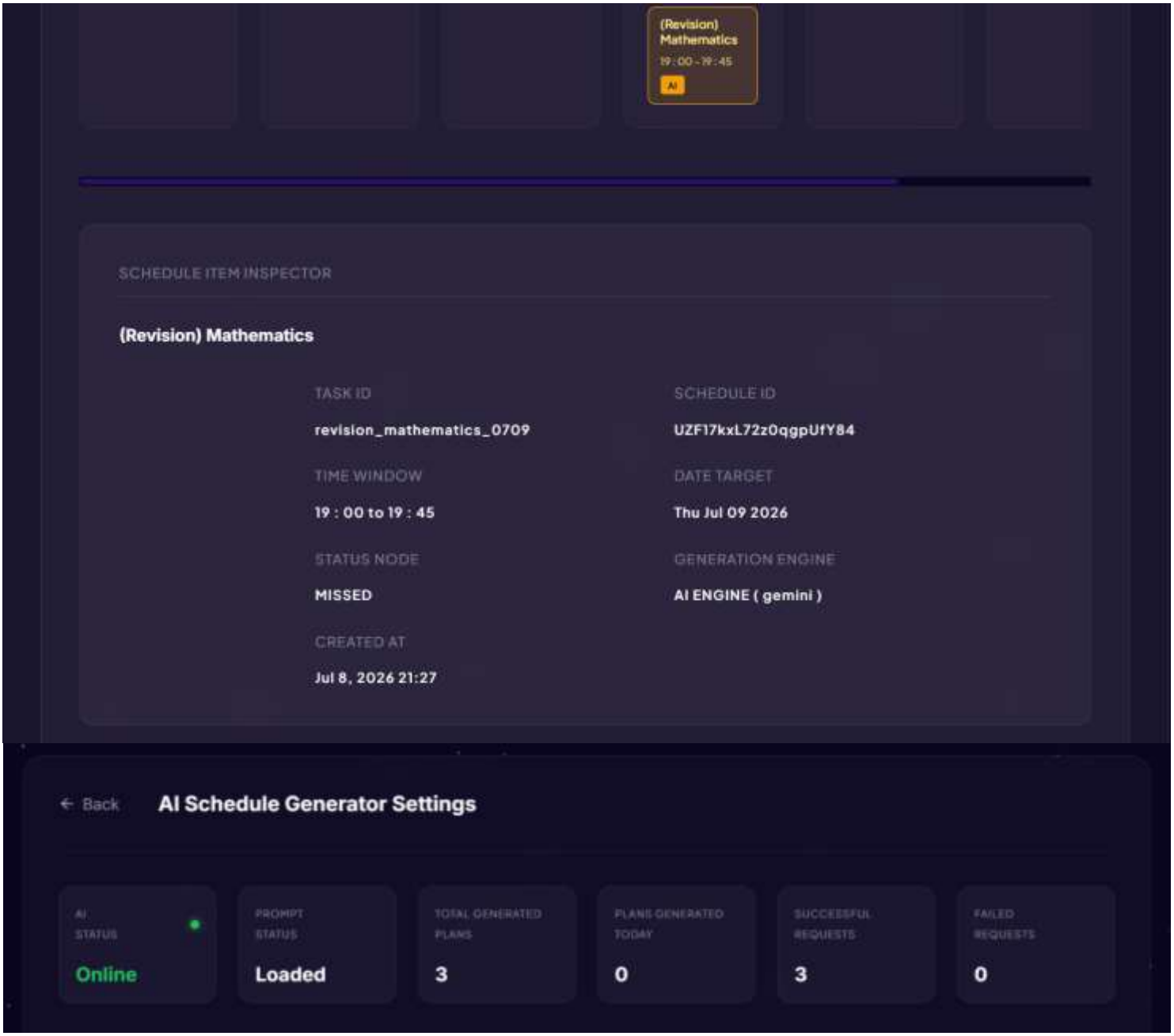


Figure 17: StudyNexus AI Schedules Control Page displaying AI prompt configurations, generated schedules, and overall system statistics

AI Schedule Monitoring and Configuration Management :

This interface provides administrators with AI schedule monitoring and configuration management capabilities by displaying AI-generated **weekly schedule** statistics stored in Firestore, retrieving and analyzing AI-created weekly schedules to evaluate performance, and loading the active scheduling prompt for review and modification. Administrators can edit and save the Gemini AI prompt configuration back to Firestore through a secure update process, with the system performing a server-side page reload after saving to prevent duplicate submissions and refresh the interface with the latest configuration.

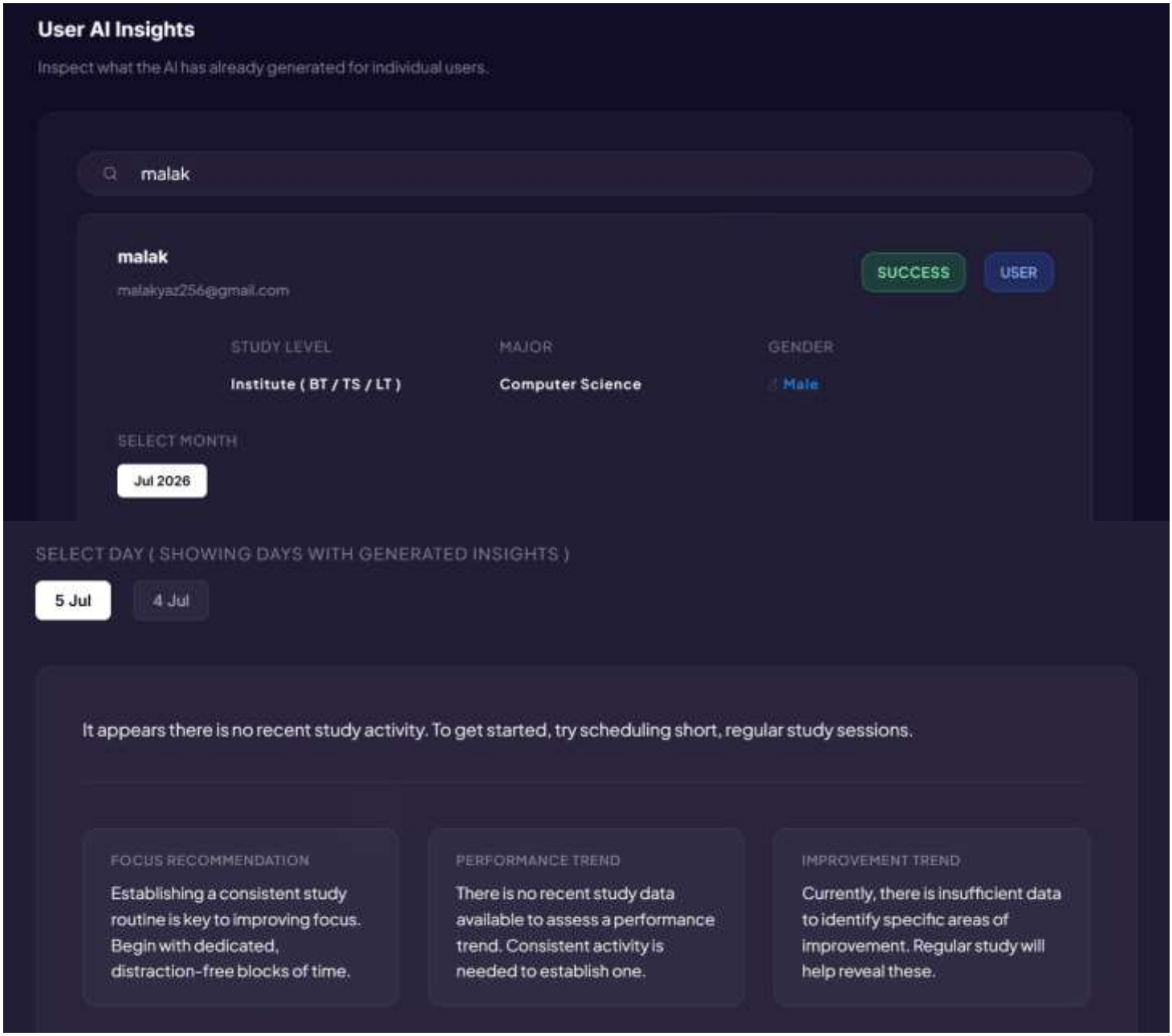


Figure 18: StudyNexus AI Insights Control Page displaying AI prompt configurations, generated insights, and overall system statistics

AI Insights Monitoring and Configuration Management:

This interface enables administrators to monitor AI-generated insight statistics stored in Firestore, review daily AI insight records, analyze recommendation quality, and manage the active Gemini AI insight generation prompt. Similar to AI schedule management, administrators can securely edit and save the prompt configuration, with the system updating Firestore and performing a server-side reload to prevent duplicate submissions. Unlike schedules, AI insights focus on evaluating daily AI-generated recommendations, performance trends, and personalized learning analysis.

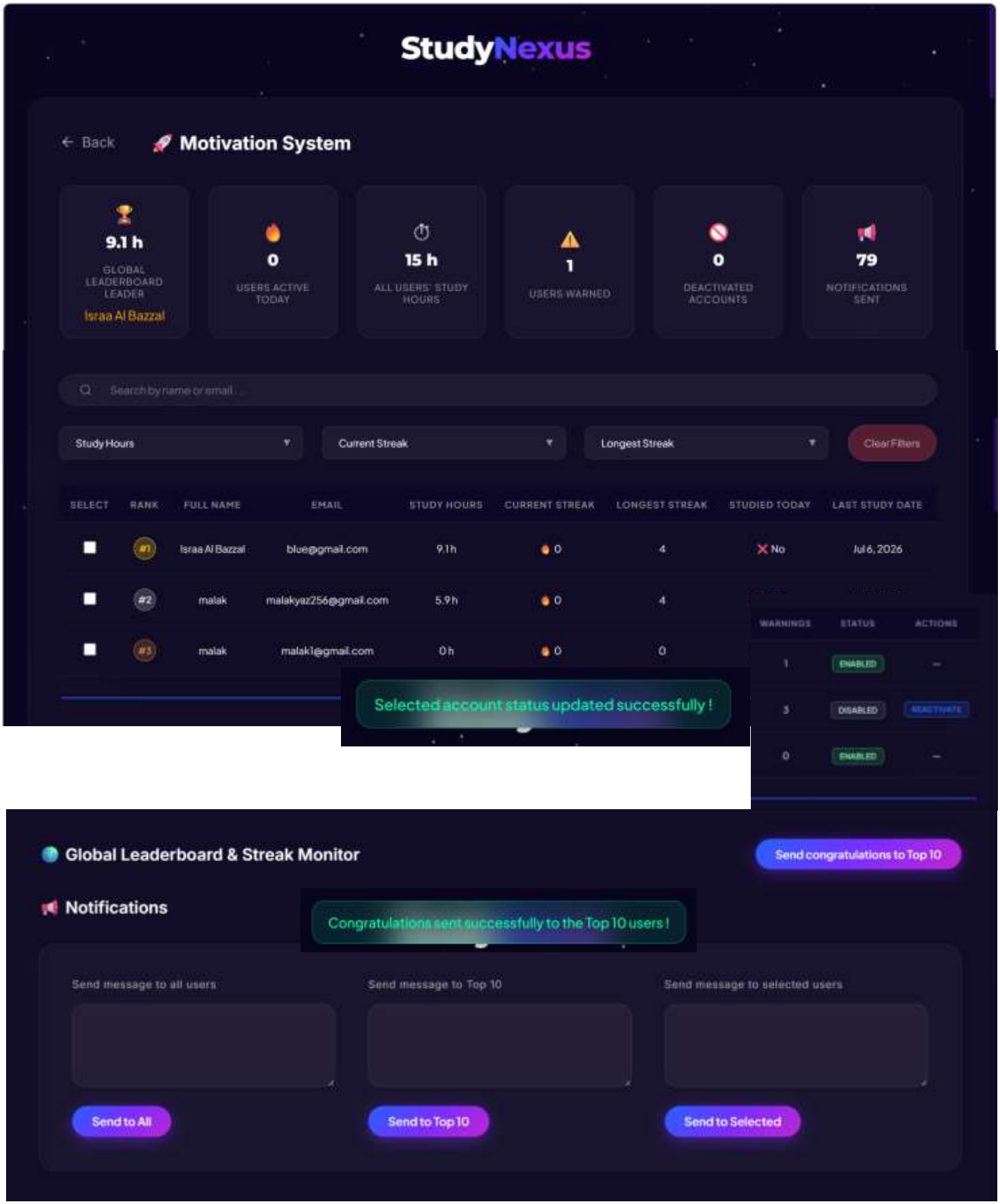


Figure 19: StudyNexus Motivation System dashboard displaying user statistics, leaderboard rankings, streak tracking, and admin notifications

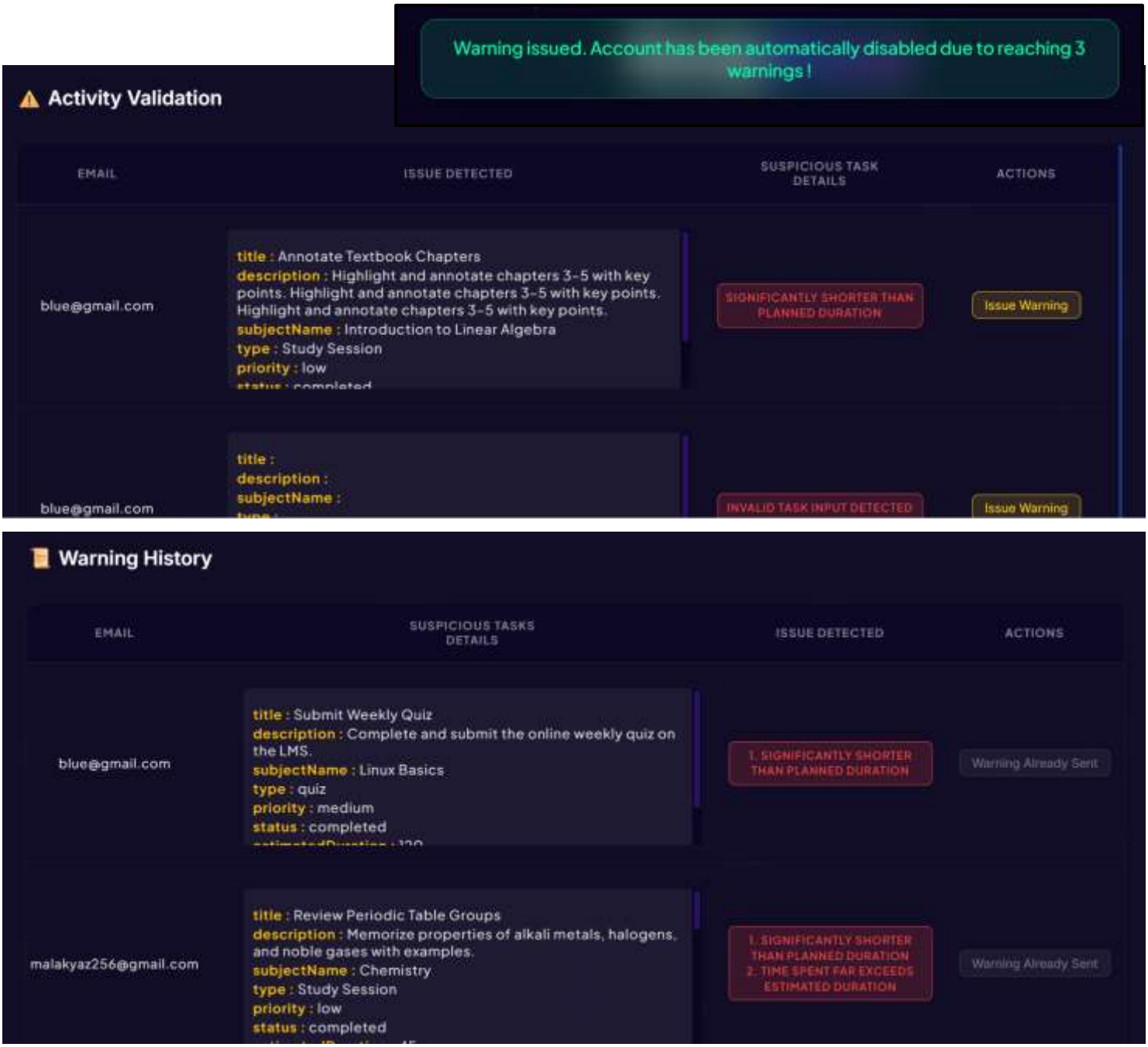


Figure 20: Activity validation and warning history sections showing detected issues and administrator warning records

Dashboard Analytics and Administrative Controls:

This interface provides administrators with a centralized monitoring dashboard displaying real-time statistics, including study activity, account status, suspicious activity detection, and warning records. It organizes user insights through global leaderboards, activity validation queues, and warning history logs to support effective management. Administrators can manage account access, handle suspicious behavior, issue or reset warnings, and send targeted notifications or motivational messages to individual users, top performers, selected groups, or the entire user base.

2. Mobile App Testing

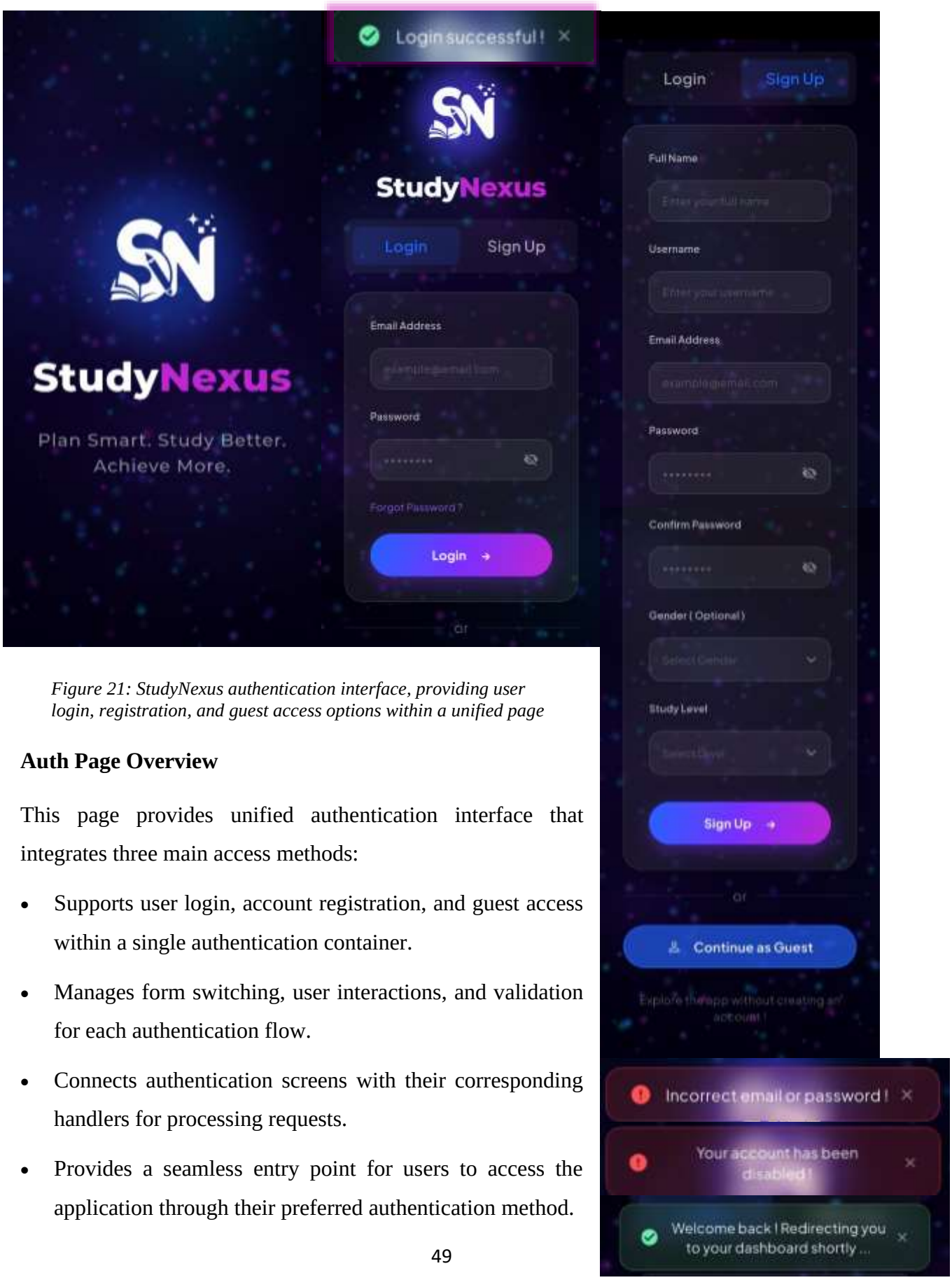


Figure 21: StudyNexus authentication interface, providing user login, registration, and guest access options within a unified page

Auth Page Overview

This page provides unified authentication interface that integrates three main access methods:

- Supports user login, account registration, and guest access within a single authentication container.
- Manages form switching, user interactions, and validation for each authentication flow.
- Connects authentication screens with their corresponding handlers for processing requests.
- Provides a seamless entry point for users to access the application through their preferred authentication method.

Authentication System Overview:

The authentication module provides a complete user access management system by integrating Flutter with Firebase Authentication, Cloud Firestore, and Realtime Database. It supports secure login, account registration, and guest account creation with validation and permission checks. The system verifies user roles and account status, blocking disabled accounts and restricting unauthorized access. User profiles, login activity, and registration data are securely stored and synchronized with Firebase services. Guest accounts are automatically initialized with unique identifiers, profile information, and presence tracking. The module also provides user-friendly feedback, loading states, error handling, and smooth navigation after successful authentication.

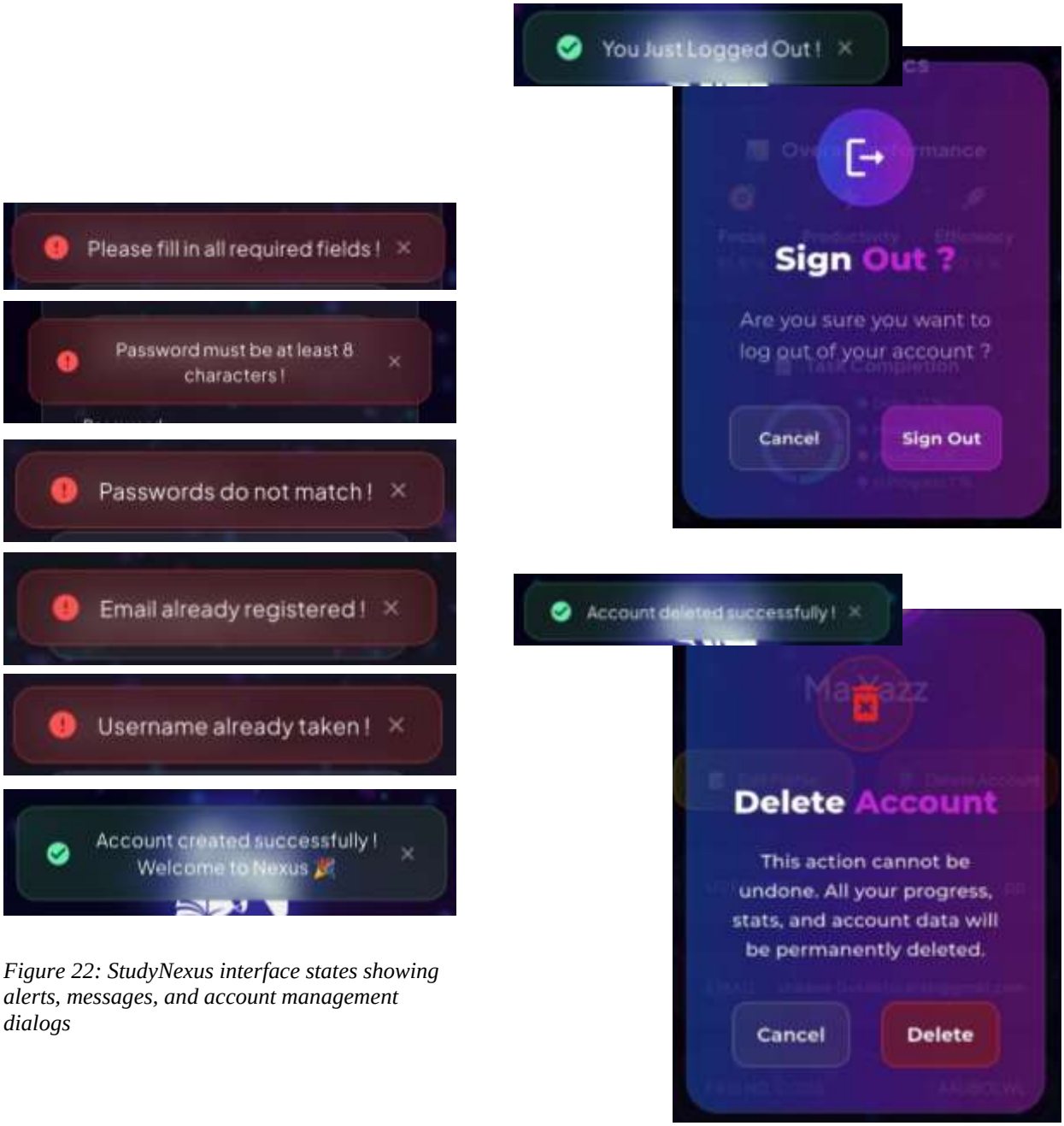


Figure 22: StudyNexus interface states showing alerts, messages, and account management dialogs

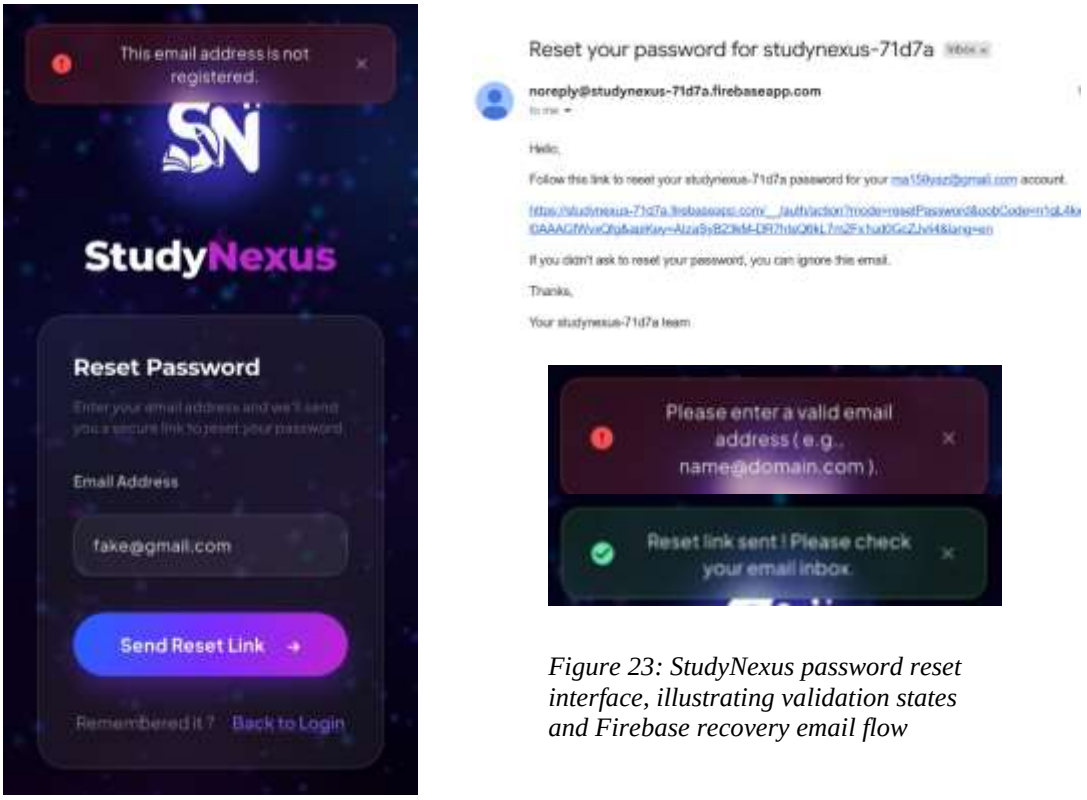


Figure 23: StudyNexus password reset interface, illustrating validation states and Firebase recovery email flow



Figure 24: StudyNexus Home Dashboard management interface, featuring user data synchronization, study metrics, notifications, weekly study hours tracking, and friend leaderboard rankings

Home Dashboard Management

This interface serves as the central dashboard controller by synchronizing user profile, study statistics, and notifications with Firestore in real time to keep displayed information up to date. It automatically calculates study metrics, performance analytics, streaks, and achievement progress in the background while maintaining data consistency by updating overdue tasks, validating schedule statuses, and tracking subject study recency. The dashboard also processes weekly leaderboard rankings among connected friends, detects rank changes, and generates notifications accordingly. Additionally, it securely manages user logout by updating presence status, recording activity logs, clearing the authenticated session, and redirecting users to the application login page.

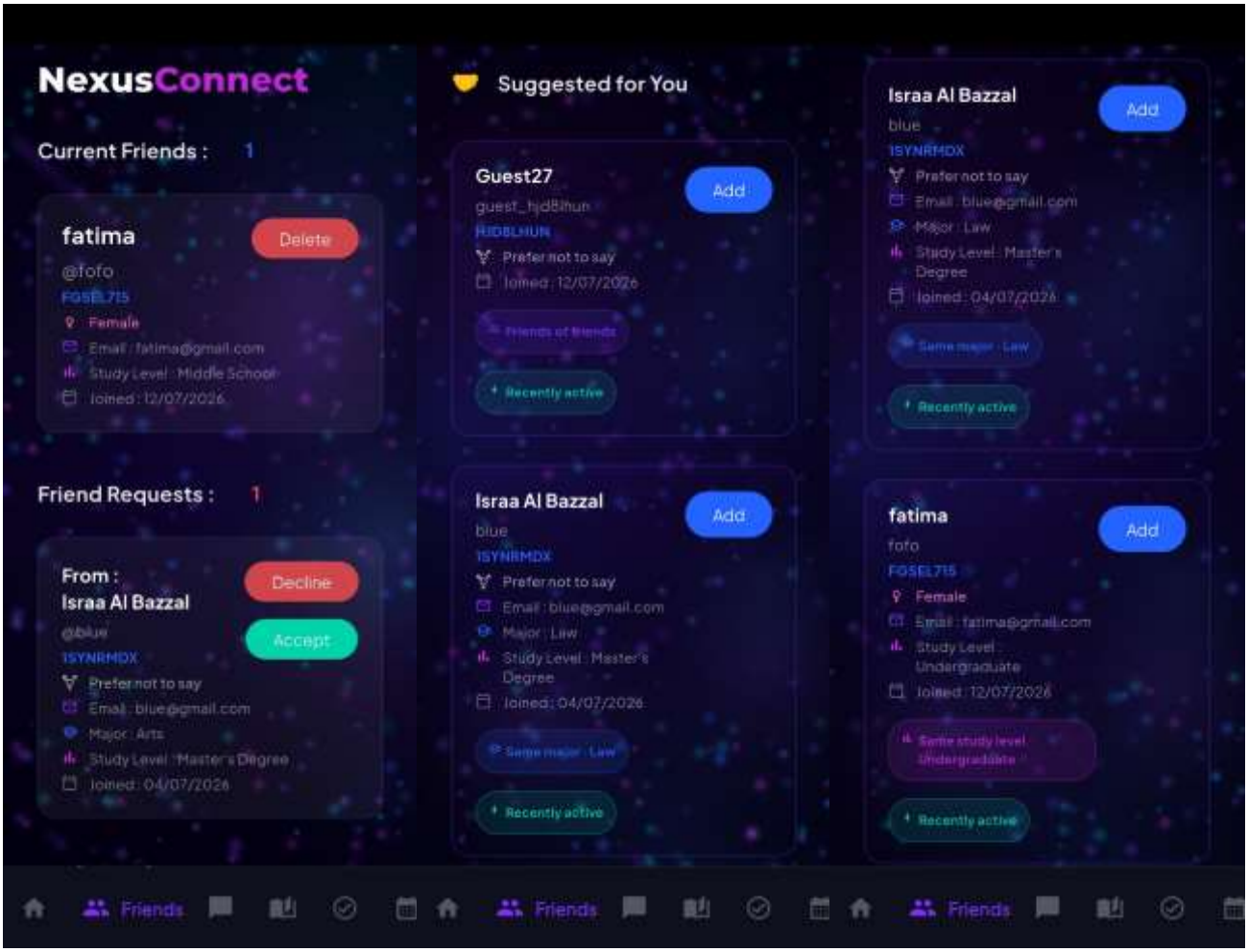


Figure 25: NexusConnect interface displaying user profiles, friends, friend requests, and user suggestions

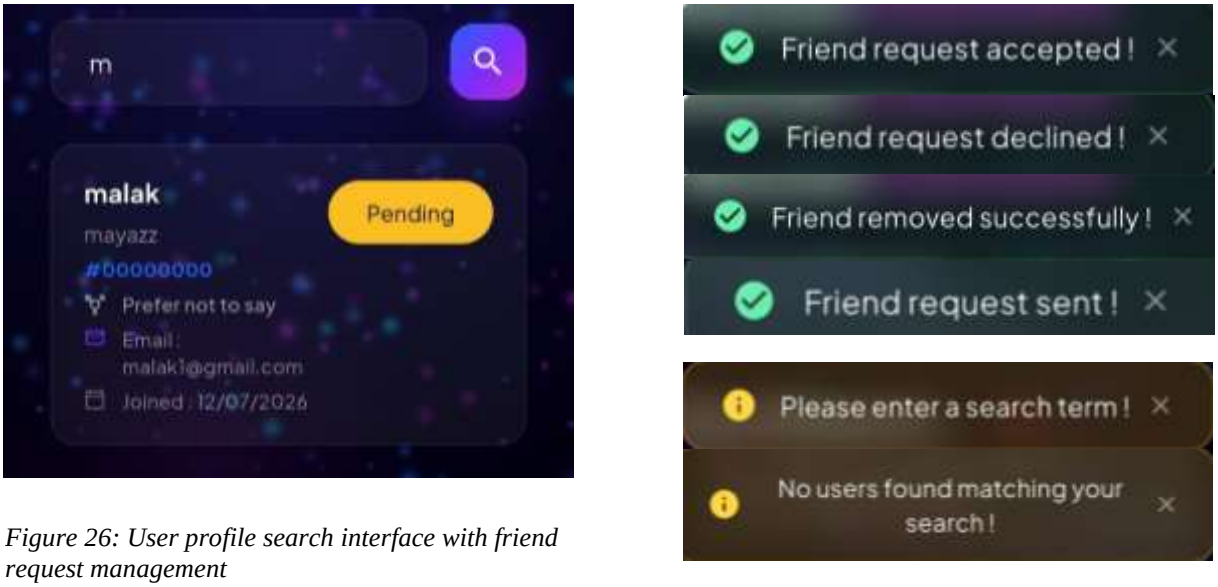


Figure 26: User profile search interface with friend request management

- **Friends Management:**

  Enables users to manage social connections through real-time friend requests, accepted friendships, and relationship updates synchronized with Firebase Firestore.

- **Friend Discovery:**

  Generates personalized friend recommendations using a weighted scoring algorithm based on mutual friends, academic information, and recent user activity while excluding existing relationships and administrators.

- **User Search:**

  Supports multi-field user discovery by searching names, usernames, emails, friend codes, majors, and study levels, with duplicate filtering and pending request detection.

- **Relationship Management:**

  Allows users to send, accept, decline, and remove friendships while preventing duplicate or conflicting friend requests.

- **Notification System:**

  Automatically generates in-app notifications for friendship events, including new friend requests, accepted requests, and other relationship updates.

- **Real-Time Synchronization:**

  Continuously retrieves and enriches friendship and user profile data from Firestore, ensuring an accurate and responsive social experience.

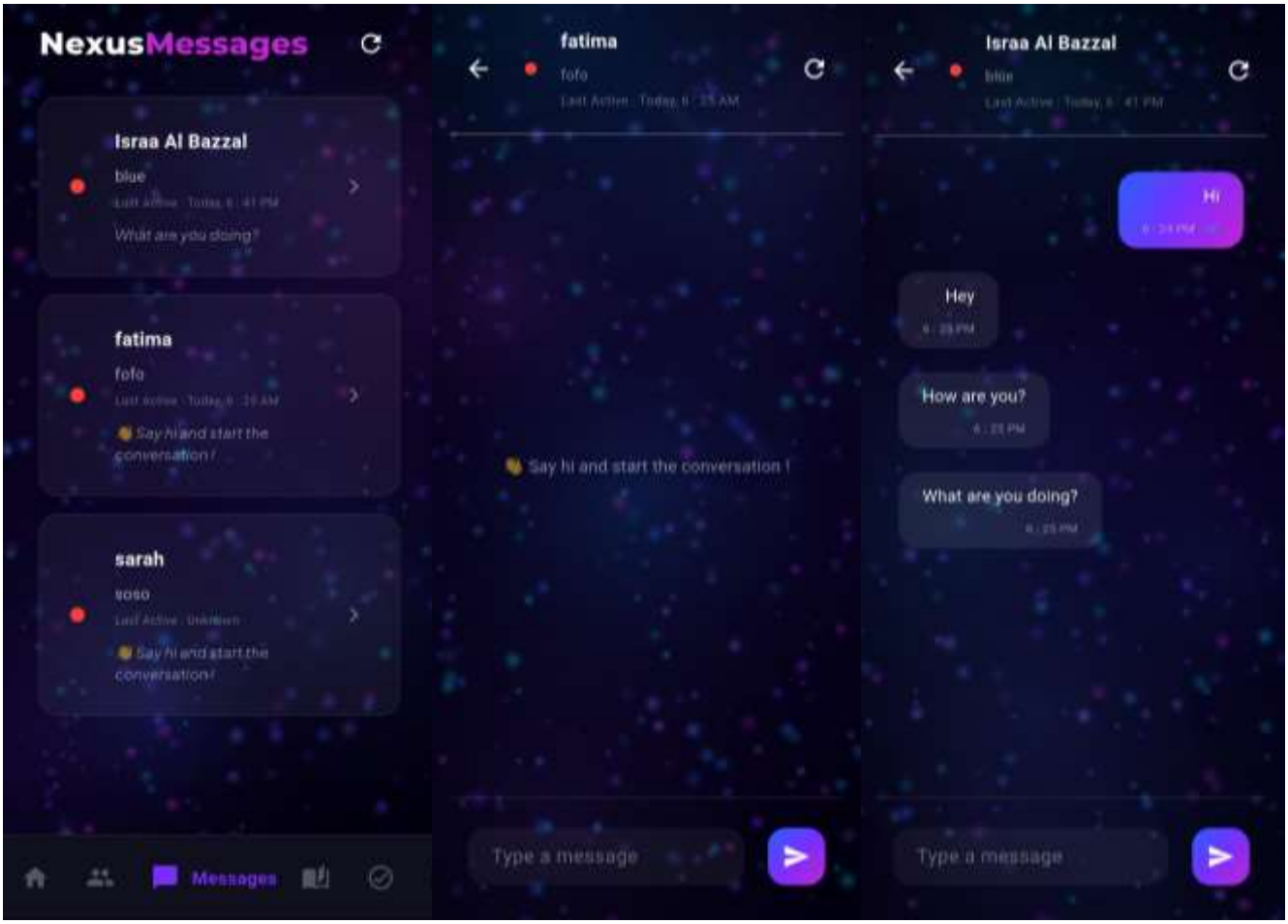


Figure 27: NexusMessages messaging interface showing conversation list, empty chat, and active conversation screens

Messages View: Displays all active conversations by retrieving accepted friendships, loading participant profiles, fetching the latest message and unread counts, sorting chats by recent activity, and automatically refreshing conversation data using adaptive polling intervals based on user presence.

Chat View: Manages individual conversations by synchronizing messages through real-time streaming when both users are online or adaptive polling when offline, automatically marking received messages as read, sending new messages with secure Firestore timestamps, logging messaging activities, and maintaining the chat view at the latest message through automatic scrolling.

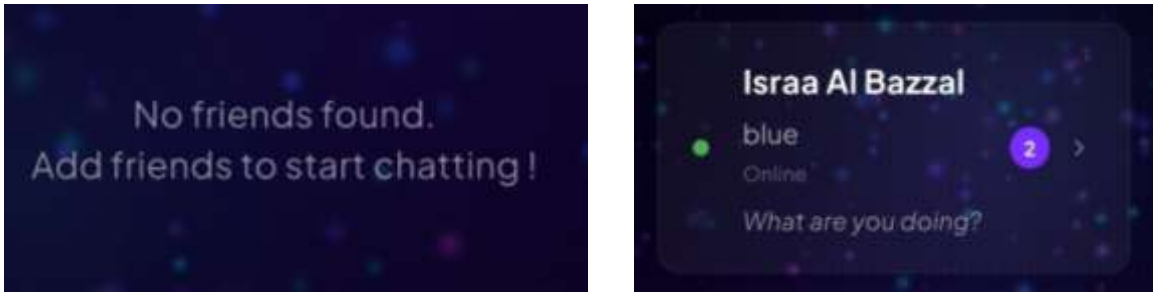


Figure 28: Chat module showing an empty state and a friend with unread messages

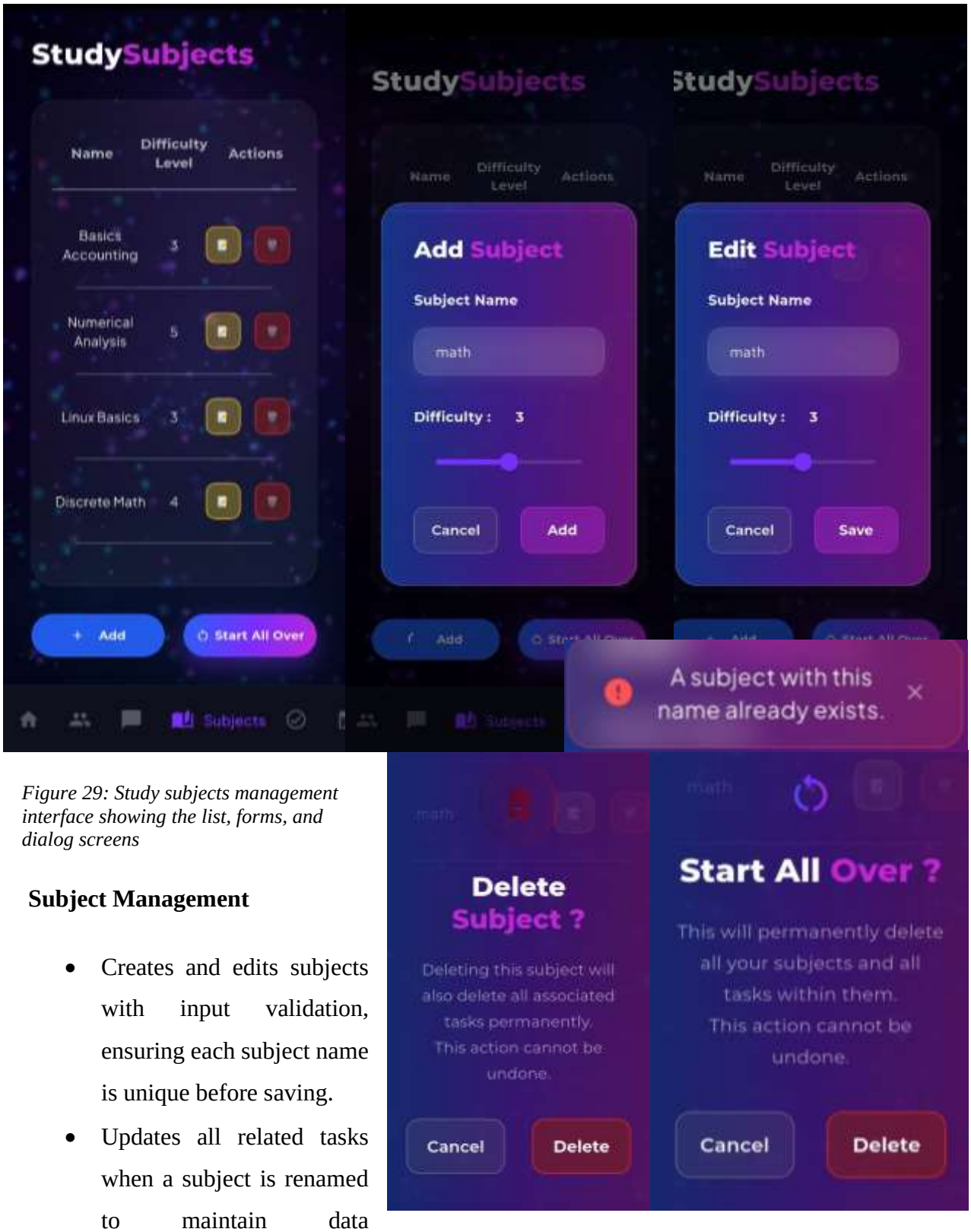


Figure 29: Study subjects management interface showing the list, forms, and dialog screens

Subject Management

- Creates and edits subjects with input validation, ensuring each subject name is unique before saving.
- Updates all related tasks when a subject is renamed to maintain data consistency.
- Deletes subjects using atomic batch operations to remove associated tasks while preserving database integrity.
- Supports bulk progress reset by removing all subjects and tasks after user confirmation.
- Provides immediate feedback through toast notifications and records subject management activities for tracking.

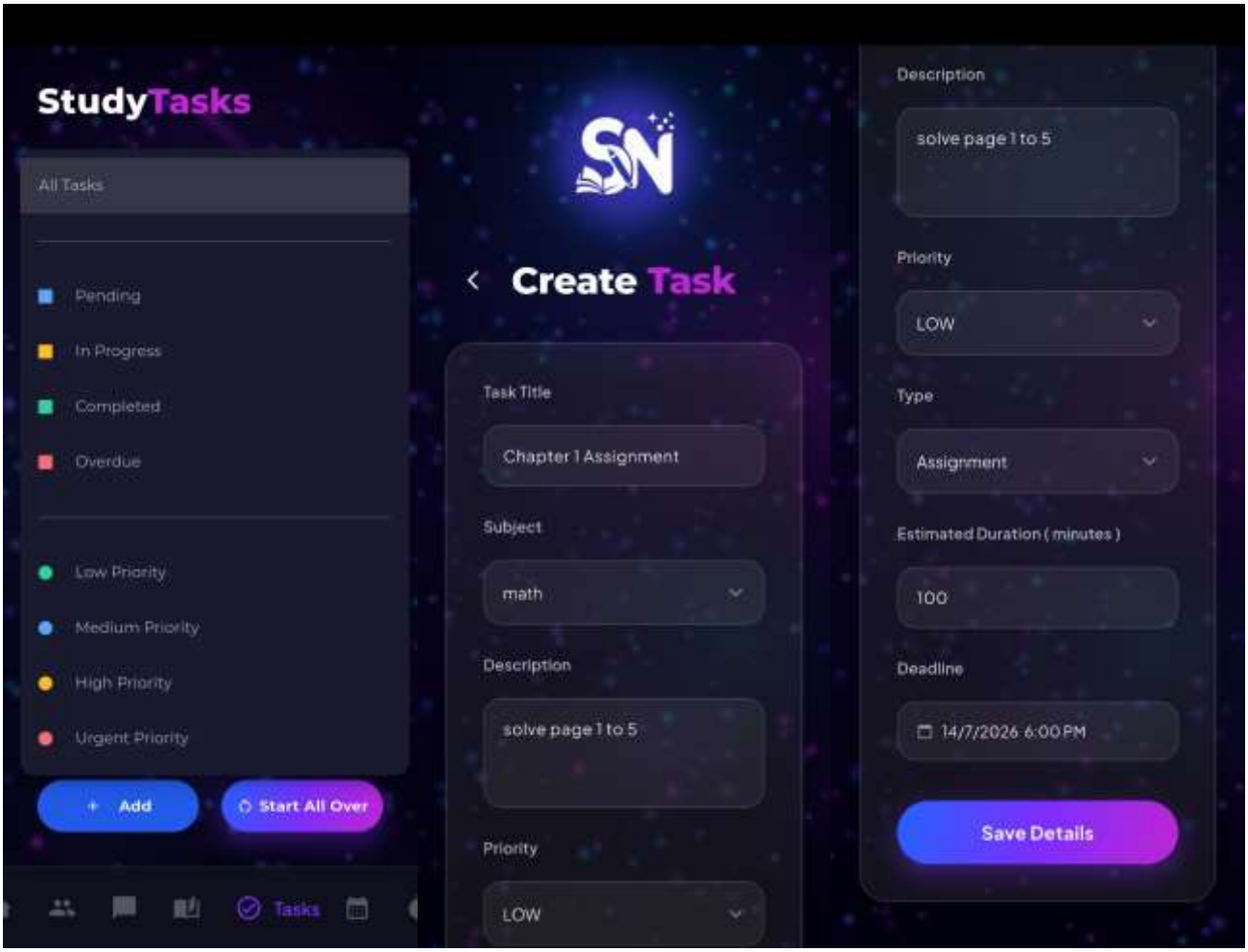


Figure 30: StudyNexus task management interface, featuring the task dashboard with priority and status filters and the task creation form

Task Management Module Overview

- Real-time task management with filtering, searching, sorting, and selection.
- Task creation and editing with comprehensive validation and duplicate prevention.
- Detailed task view with automatic data refresh.
- Atomic batch deletion of tasks and associated schedules.
- Dynamic task status calculation for accurate progress tracking.
- Loading states, rapid-action prevention, and proper resource management for stability.

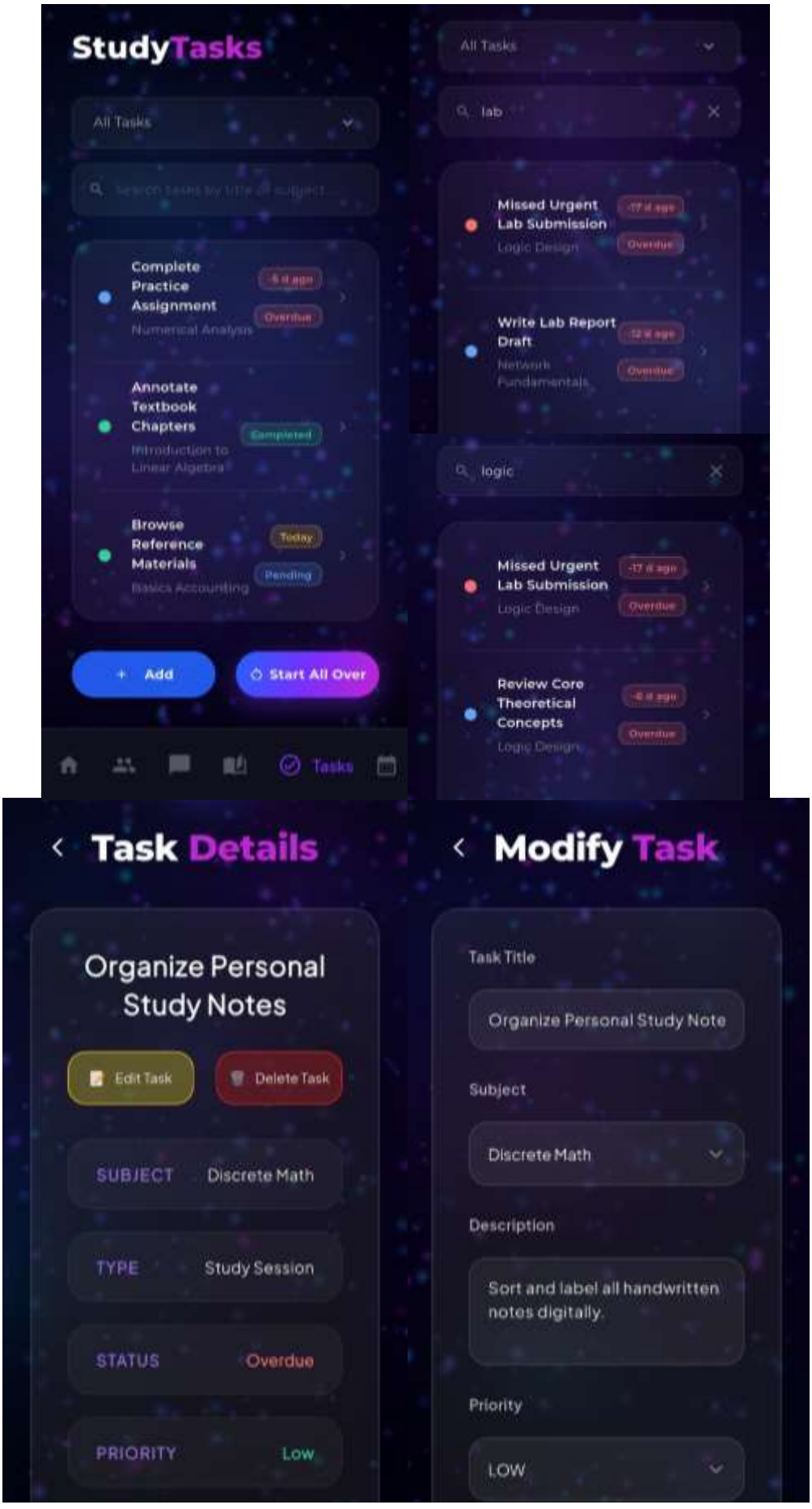


Figure 31: StudyNexus task management interface showing the task dashboard with filtering options, task details, and task editing screens

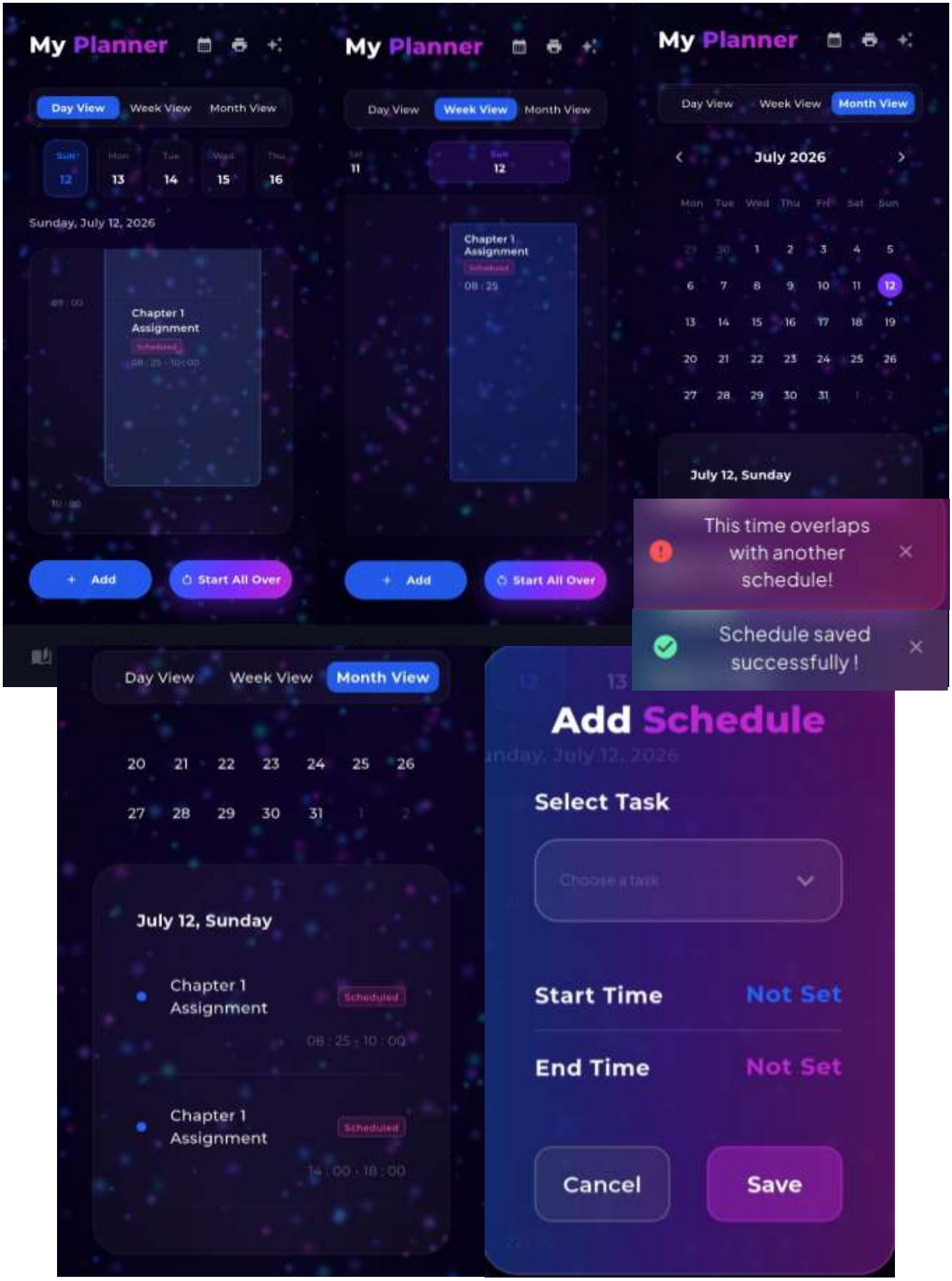


Figure 32: StudyNexus planner interface showcasing day, week, and month calendar views, schedule creation, and validation toasts

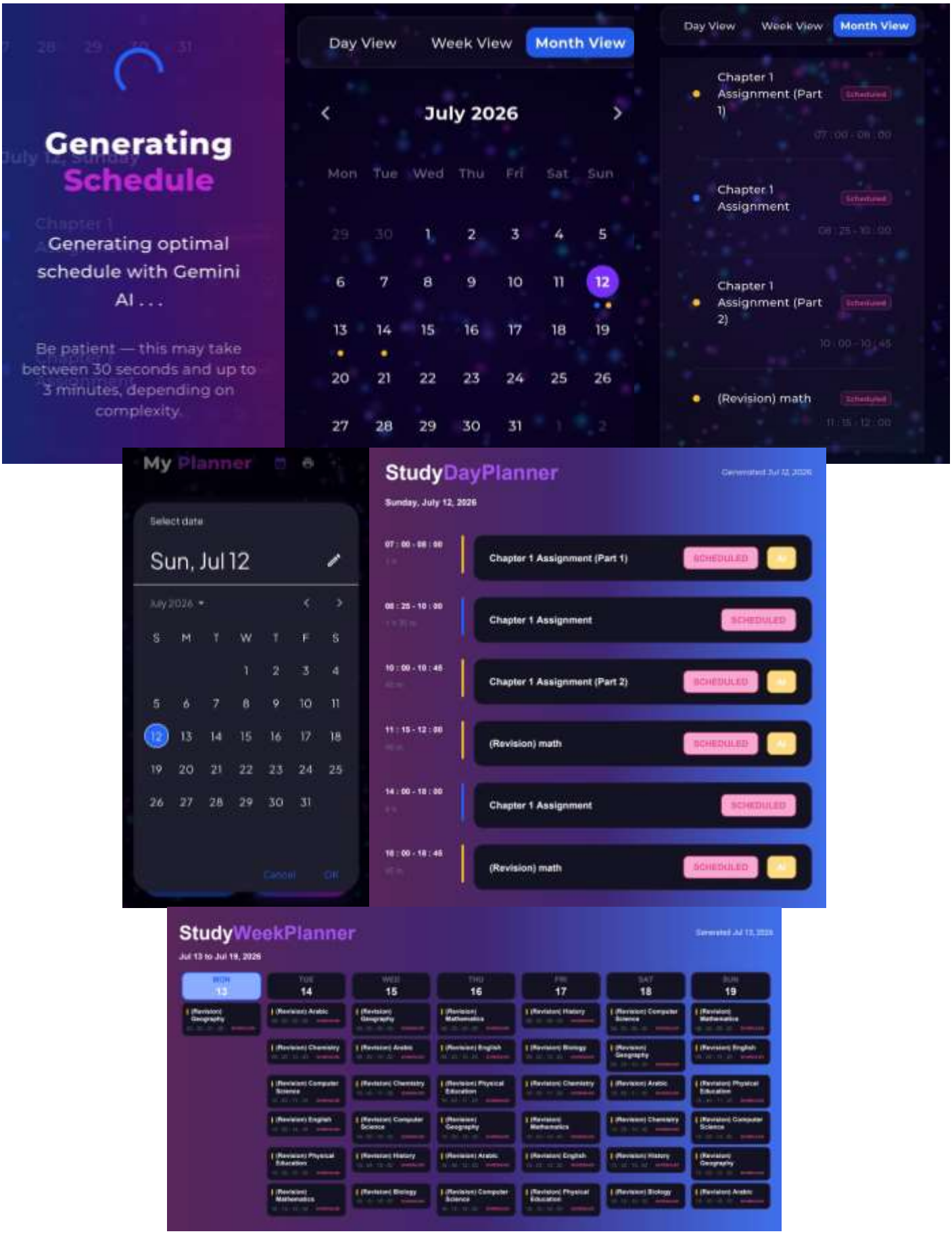


Figure 33: StudyNexus smart planner interface showcasing AI-generated schedules and PDF schedule exports

Planner Management Module Overview

- Displays schedules in real time using Firestore streams with automatic synchronization across Day, Week, and Month planner views.
- Supports creating, editing, and deleting schedules with comprehensive validation, including required field checks, valid time ranges, and prevention of overlapping schedules through Firestore transactions.
- Integrates with AI schedule generation while preventing duplicate AI-generated schedules for the same week and recording generation activities.
- Exports Day and Week schedules as PDF and allows resetting all schedules through atomic Firestore batch deletion.
- Provides a modern planner interface with multiple calendar views, color-coded schedule indicators, gradient styling, and floating toast notifications.
- Maintains data integrity through Firestore transactions and activity logging for all successful and failed schedule operations.

The **AI-Generated Schedule** module automates personalized study planning by analyzing the user's subjects, tasks, exams, study history, and available time slots before generating an optimized weekly schedule using the Gemini AI model. The generated schedule is validated to ensure it meets study duration, time, and conflict constraints, then saved to Firestore using batch operations. Throughout the process, the system includes authentication checks, subject progress updates, retry mechanisms, error handling, and telemetry logging to ensure reliable schedule generation.

The **Planner Pdf Export** service generates professional daily and weekly planner PDFs that match the application's dark-themed design. It organizes schedules into clear timeline or calendar layouts, highlights task status and AI-generated sessions, supports native PDF sharing, and records all export activities through the activity logging service. By encapsulating document creation, formatting, sharing, and logging within a reusable service, it provides a consistent and reliable planner export experience.

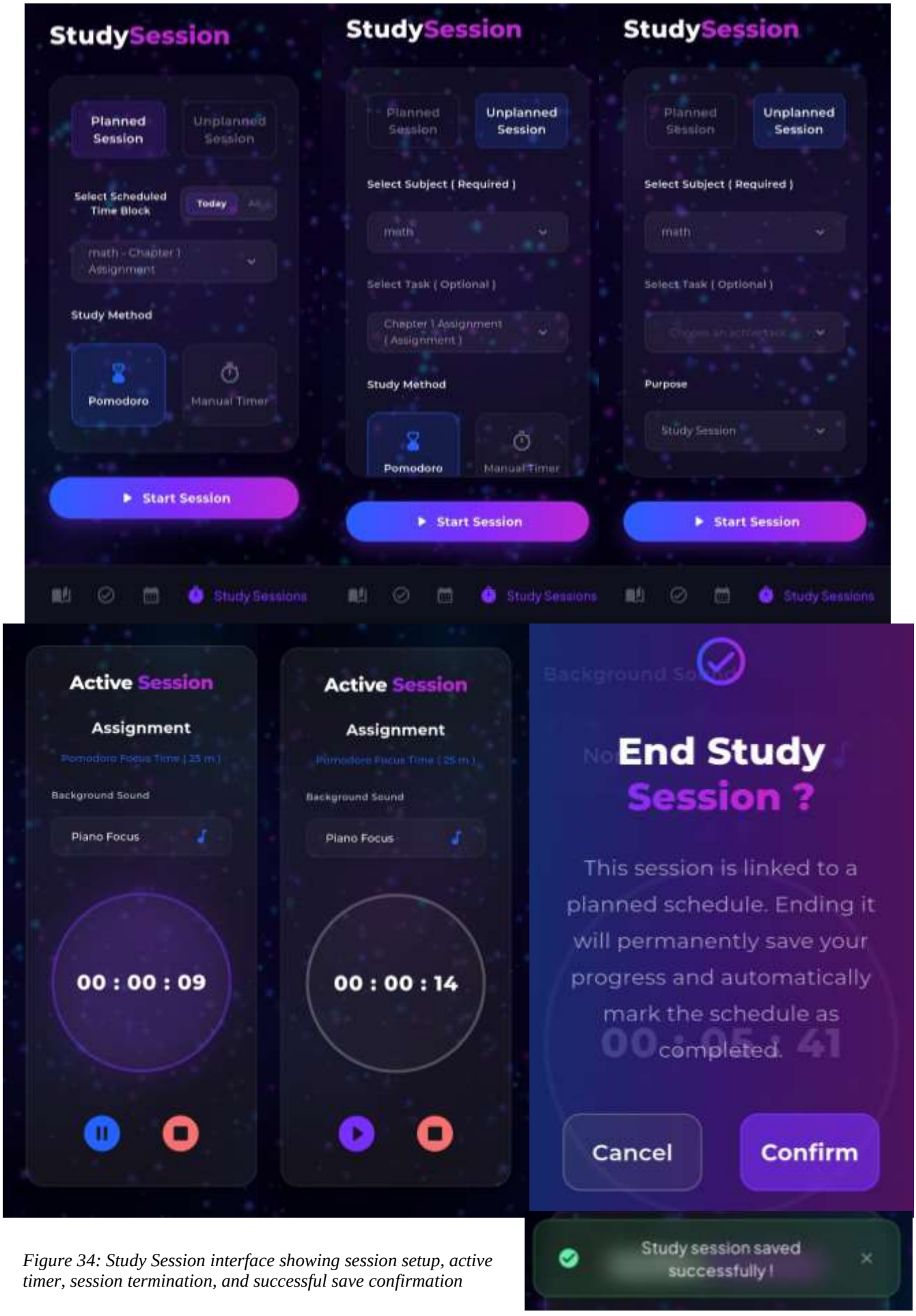


Figure 34: Study Session interface showing session setup, active timer, session termination, and successful save confirmation

Study Session Setup

Configure and launch planned or unplanned study sessions by loading subjects, tasks, and schedules from Firestore after user authentication. The setup prepares all required session data, supports flexible scheduling options, and automatically refreshes application data after each completed session.

- Authenticate users before starting sessions.
- Load and organize schedules, subjects, and tasks.
- Support planned and unplanned study sessions.
- Generate session configuration automatically.
- Refresh Firestore data after session completion.

Study Session Timer

Manage the complete study session lifecycle with real-time tracking, Pomodoro automation, break management, audio controls, and performance evaluation. Upon completion, the system saves all session data and related updates using atomic Firestore operations before launching the reflection screen.

- Track focus time, total time, and breaks.
- Support automated Pomodoro cycles.
- Manage study audio and notifications.
- Update tasks and schedules automatically.
- Save all changes through a single Firestore batch.
- Open the reflection view after completion.

Session Reflection

Collect post-session feedback by allowing users to rate their study experience and update task completion. Reflection data and task progress are saved together using atomic Firestore updates while preventing duplicate submissions.

- Record feeling, difficulty, and task completion.
- Update task status automatically.
- Save reflection and task data atomically.
- Handle submission success and error states.

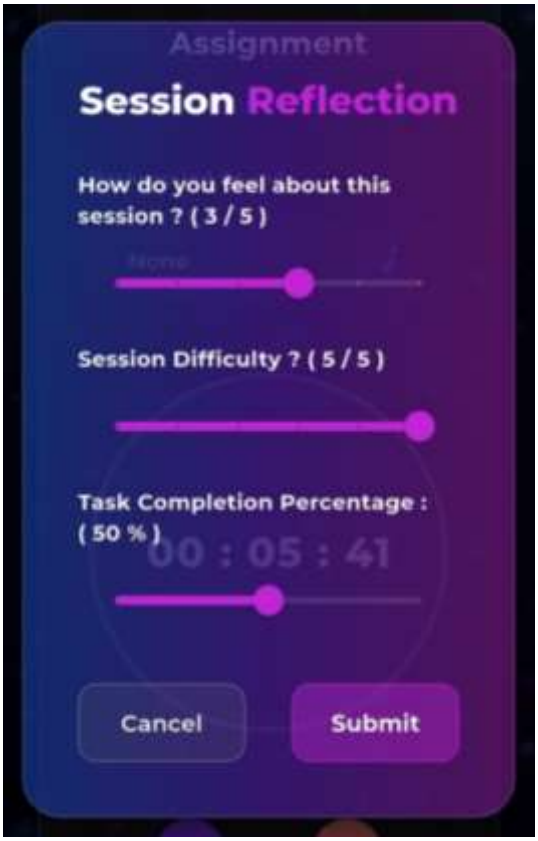


Figure 35: Session reflection interface with optional task completion tracking



Figure 36: StudyNexus AI insights dashboard analyzing study habits, performance trends, and personalized recommendations

AIView (Frontend)

- Displays AI-generated insights using a **cache-first** loading strategy.
- Verifies user authentication before retrieving insights.
- Checks the latest Firestore insight document and generates new insights only when:
    - No valid insight exists.
    - The timestamp is missing or invalid.
    - The cached insight is **7 days or older**.
- Reuses cached insights whenever they remain within the 7-day freshness window.
- Normalizes retrieved data by removing metadata, converting values to strings, and discarding empty entries.
- Displays only insight categories containing meaningful recommendations as cards with an icon, title, and recommendation.

Shows an empty state encouraging users to log additional study activity when no valid insights are available.

AIInsightsService (Backend)

- Serves as the backend orchestration layer for AI-powered academic coaching in StudyNexus.
- Aggregates up to **60 days** of Firestore data, including study sessions, tasks, subjects, and behavioral analytics.
- Processes the data across **7-, 30-, and 60-day** analysis windows to identify consistency, focus decline, performance trends, and burnout risk.
- Builds a compact prompt and sends it to **Gemini (gemini-2.5-flash)** using a dynamically loaded system instruction with a built-in fallback.
- Enforces a strict **JSON-only** response format.

- Uses **retry logic with exponential backoff (2s, 4s, 8s)** to recover from transient failures.
- Cleans and validates the returned JSON, removing formatting artifacts before processing.
- Accepts only a whitelist of approximately **18 predefined insight fields** (e.g., *aiRecommendation*, *burnoutRisk*, *studyTrend*) to prevent malformed or injected data.
- Stores one validated insight document per day in **users/{uid}/aiInsights/{YYYY-MM-DD}**.
- Logs successful operations through **ActivityLogger**, while failures are recorded in an audit collection and safely return null.



Figure 37: NexusAnalytics dashboard interfaces displaying study tracking metrics, including consistency, subject performance, and learning trend analysis



Figure 38: StudyNexus analytics page displaying study patterns, time distribution, productivity metrics, and focus behavior insights

Performance & Analytics Dashboard Summary

- **Data Processing Workflow**
    - ✓ Retrieves study data from Firestore collections including **study sessions, tasks, subjects, and user records**.
    - ✓ Cleans and validates data by handling missing values, invalid timestamps, and incomplete records safely.
    - ✓ The dashboard displays the final processed analytics through a single data source without performing calculations itself.
- **Analytics Helper Services**
    - **Consistency Service**
        - ✓ Calculates consistency scores based on completed tasks compared to total tasks.
        - ✓ Generates study streak information and calendar heatmap activity levels.
    - **Deep Learning Trends Analyzer**
        - ✓ Analyzes study sessions by time, duration, and focus levels.
        - ✓ Identifies patterns such as preferred study periods and productivity changes during longer sessions.
    - **Focus Behavior Analytics Service**
        - ✓ Evaluates focus performance using completion rates, interruptions, and overdue task patterns.
        - ✓ Produces behavior-based insights such as focus improvement or interruption risks.
    - **Subject Intelligence Service**

- ✓ Compares study time distribution with productivity results for each subject.
- ✓ Identifies high-efficiency subjects and areas needing improvement.

- **Time Efficiency Analysis**
  - ✓ Compares productive study time by analyzing session duration and break periods.
  - ✓ Detects whether shorter or longer sessions provide better performance.

- **Calculated Insights**
  - ✓ Insight messages displayed to users are **rule-based calculations**, not AI-generated responses.
  - ✓ Predefined conditions and thresholds convert calculated metrics into personalized messages.

- **Result**
  - Converts raw academic activity data into meaningful visual analytics.
  - Provides a personalized dashboard experience while maintaining deterministic and reliable calculations.

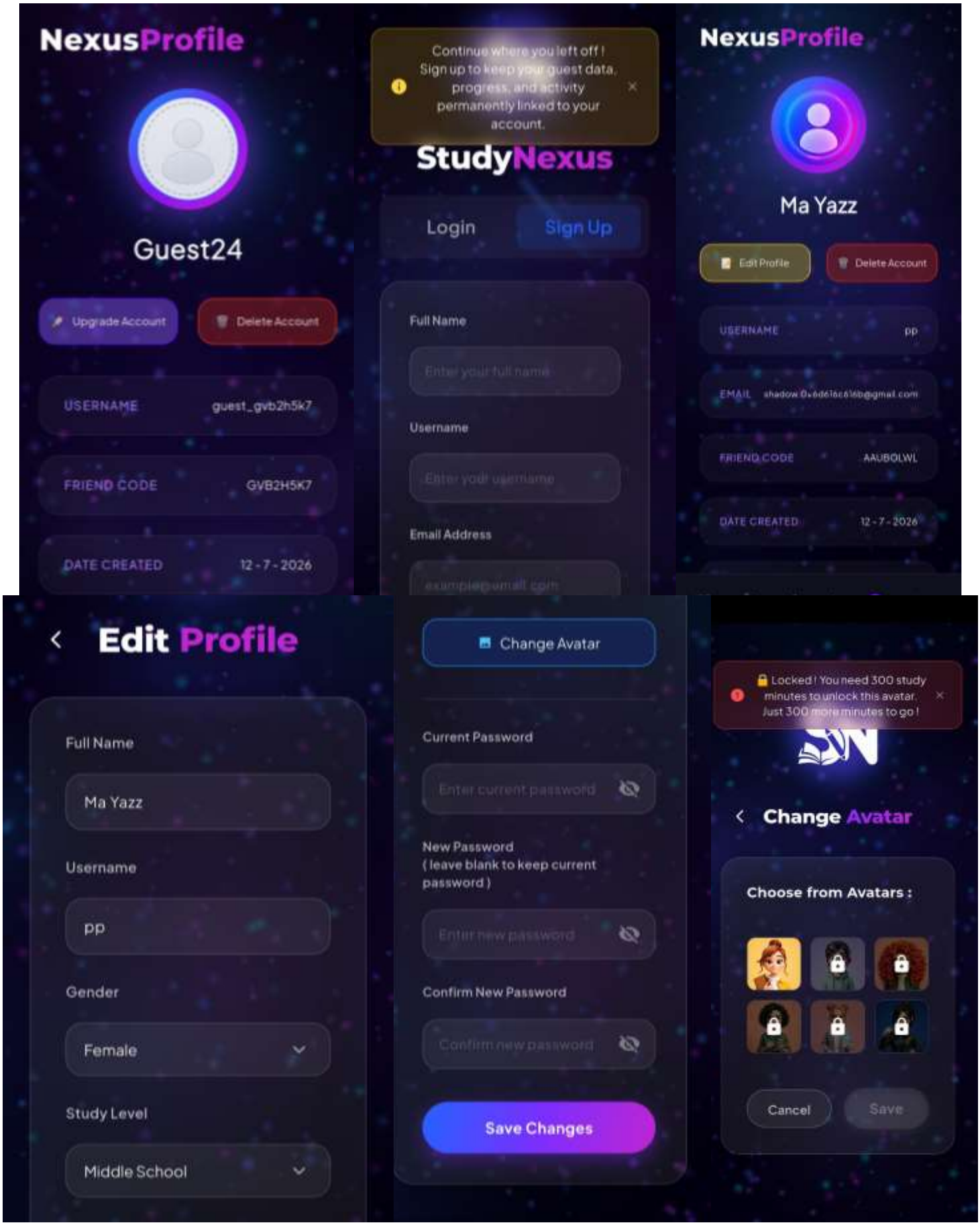


Figure 39: User profile and account management interfaces in the StudyNexus application, comparing guest and registered user accounts, including sign-up with guest account linking, profile editing, password updates, and avatar customization

Profile & Avatar Management System

1. Guest vs. Registered Accounts

- Guest and registered users see **different avatar sets** — avatars are filtered by the user's selected gender category (Female/Male/Neutral), and guest profiles display a limited/placeholder view of profile fields until upgraded.
- Guests are shown **Upgrade** and **Delete** options; registered users are shown **Edit** and **Delete** options instead.
- A user **cannot create a new guest account** while an existing guest account is still active — the old guest profile must be deleted first before a new guest session can be started.

2. Guest Account Upgrade

- startGuestUpgrade() logs the upgrade attempt via ActivityLogger, then routes to AuthPage carrying the **existing guest UID** as a parameter.
- Signup runs in "guest-linking" mode, so the new registered account is tied to the **same UID/data record** — study time, avatar progress, and profile info carry over instead of being lost.

3. Profile Editing

- EditProfilePage restores existing values (including avatar and dropdowns) on load, with "Other" major auto-detected and filled if not in the preset list.
- Validates required fields, checks username uniqueness in Firestore, and — if changing password — requires reauthentication before Firebase updates it.
- Saves updates name, username, study_level, gender, major, avatar_filename, plus an updated_at timestamp; redirects to login only if the password changed.

4. Avatar Selection & Rewards

- Avatars are grouped by gender (6 per category) and unlocked progressively based on **accumulated study minutes**

- This tiered unlock system acts as an **avatar reward mechanic** — locked avatars show the remaining minutes needed rather than being selectable.
- Saving a new avatar updates avatar_filename in Firestore, logs the action, and returns the selection to the profile screen.

5. Account Deletion

- Deletes Firestore profile data, then the Firebase Auth account, logging differently for guest vs. registered users; prompts re-authentication if Firebase requires it for security.

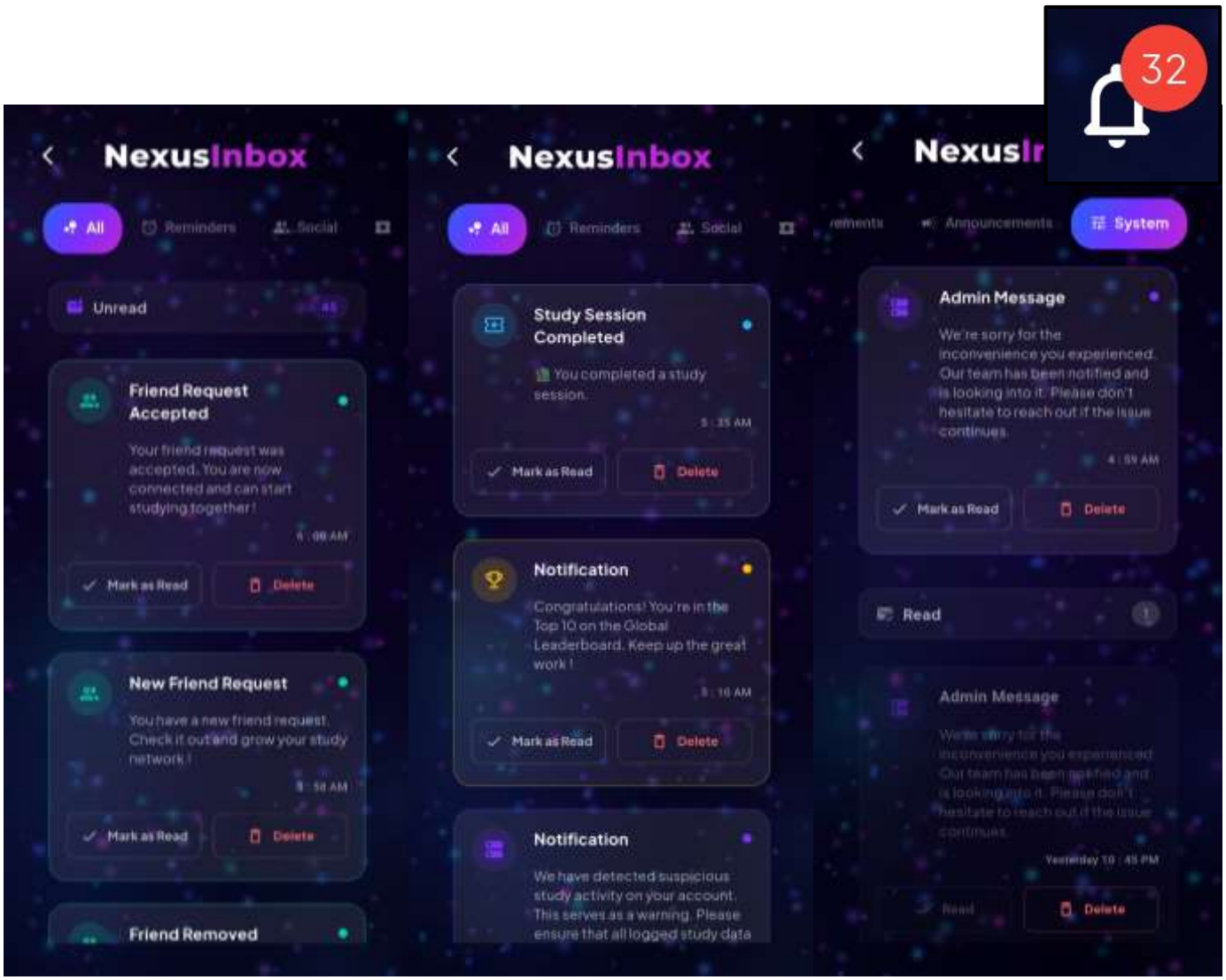


Figure 40: NexusInbox notification interface showing category filters and notification management features, including read, unread, and delete actions

Notification Handling System

- Organizes notifications into categories:
    - All, Reminder, Social, Activity, Congratulations, Broadcast, System.
- Filters notifications based on:
    - Selected category.
    - Visibility status (hidden notifications are excluded).
    - Release time (future notifications remain hidden until available).
- Sorts notifications by timestamp, showing the newest alerts first.

Display & User Interaction

- Separates notifications into:
    - Unread notifications displayed first.
    - Read notifications displayed with reduced emphasis.
- Applies category-based visual styles and formats timestamps into readable forms.
- Supports:
    - **Mark as Read:** updates isRead = true in Firestore.
    - **Hide Notification:** updates status = hidden without deleting the record.

Overall Result

- Provides a personalized notification dashboard synchronized with Firestore in real time.
- Ensures efficient notification organization while preserving notification history by hiding records instead of permanently removing them.

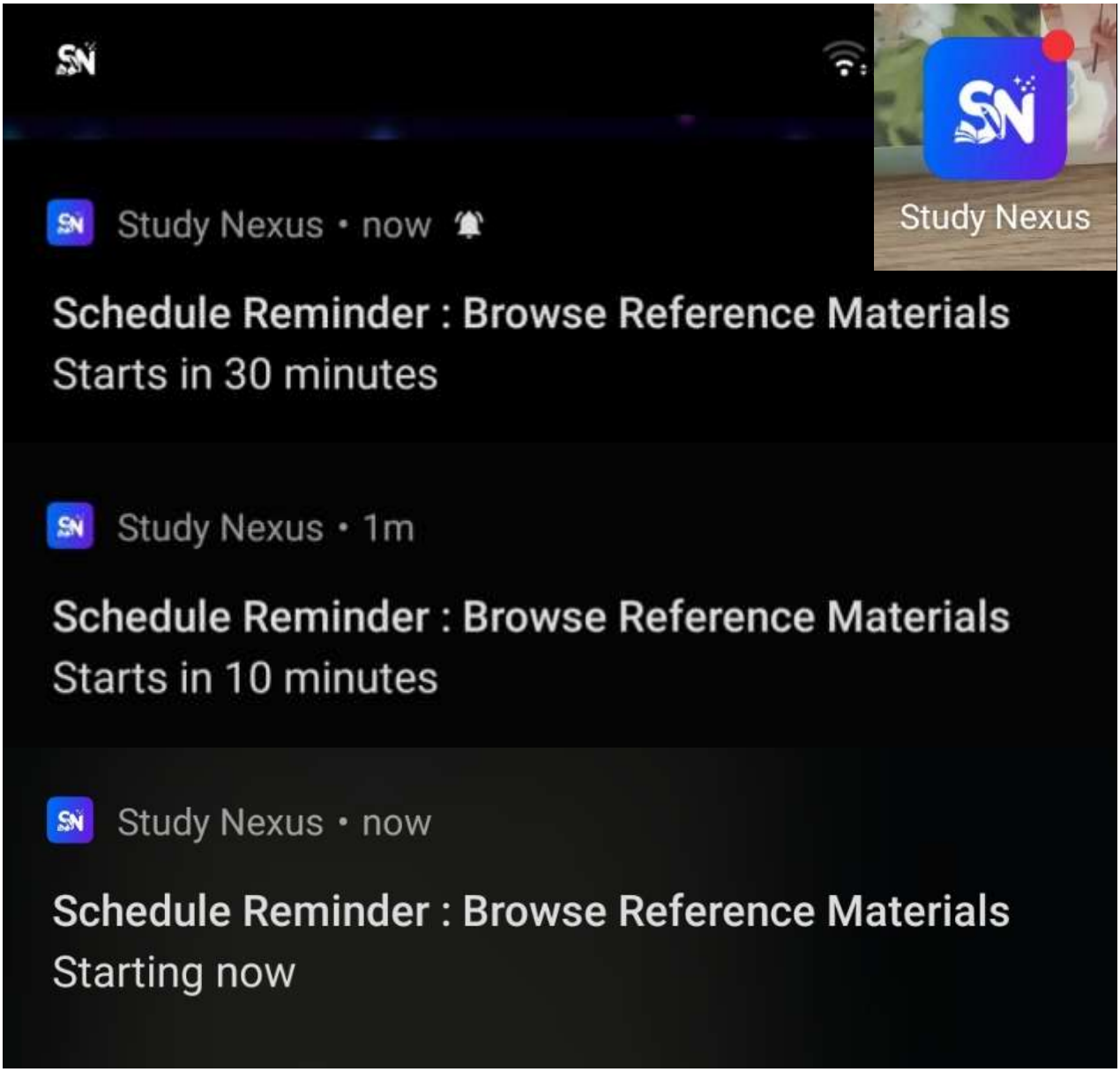


Figure 41: StudyNexus push notification interface displaying schedule reminders for upcoming and active study sessions

Notification Scheduling & Reminder System

This page manages user reminders by synchronizing study schedules with device notifications. It ensures users receive timely alerts before, during, and after study sessions, helping them stay organized and maintain their planned activities. The system provides reliable background reminders while keeping notification data synchronized with the user's schedule.

General Conclusion and Future Recommendations

StudyNexus isn't meant to be a finished product. It's more like a smart foundation for what academic management could look like if you built with AI at the center. You get study planning, productivity tracking, performance analytics, and social collaboration—all bundled together. It makes life easier for students who need help staying organized. And honestly, the platform is flexible enough to stretch way beyond what it does right now. As AI and educational tech keep improving, you can imagine a whole lineup of new features and intelligent services that make learning even better.

For this project, the essential features that actually deliver immediate value. But there's plenty of room to grow. Here are a few ideas to make StudyNexus smarter, more reliable, and more collaborative:

1. AI-Based Automatic Task Completion Evaluation

Right now, users manually type in how much of a task they finished after a study session. It's simple, but let's be real: self-reporting is pretty subjective. Sometimes people overestimate or underestimate their progress, and that throws off productivity stats and AI insights.

A smarter fix is letting AI handle the evaluation. Instead of users entering a percentage, they just type a quick summary of what they accomplished. Then, using natural language processing and big language models, the system reviews their reflection, the original task, study duration, past progress, behavior—with all the session data—and figures out the completion percentage for them.

But it doesn't just spit out a number. The AI gives a quick explanation, showing what got done and what's left to tackle. That makes the whole process transparent and a lot more honest, and it takes the guesswork out of progress tracking.

With this upgrade, productivity measures get more objective, AI recommendations get sharper, and students have a clearer sense of where they stand. Over time, the data from these sessions could even fuel more personalized study advice and spot patterns to predict future performance.

2. Collaborative AI Study Groups with Privacy Protection

Another big step forward: collaborative study groups. Imagine students teaming up for group sessions, scheduling together, supporting each other, and staying motivated. The trick is keeping everyone engaged without forcing hours-long video calls.

Instead, StudyNexus could snap random screenshots—only with explicit consent—and share them inside the group. These snapshots provide accountability that participants are actually studying, but respect privacy and cut out the stress of always being on camera.

Compared to endless streaming, this is way less intrusive. Many students don't love staying on camera, or their internet can't handle it. Random, consent-based snapshots do the job without causing discomfort. Everyone still gets the push to stay focused, but can work independently.

You could make group study even more fun by adding shared Pomodoro timers, collective stats, progress charts, motivational AI messages, and achievement badges—celebrating both solo and group wins. That positive reinforcement helps build solid study habits and a sense of support.

Final Thoughts

StudyNexus proves that mixing AI with productivity tools, study planning, analytics, and social features can genuinely lift the academic experience. The system isn't just a solution for today—it's built to expand and handle whatever tomorrow's students need, tackling the big challenges of staying organized and motivated.

As AI gets smarter, StudyNexus can transform from just a helpful platform into a full learning ecosystem—one that guides students personally, evaluates their performance fairly, encourages group collaboration, and keeps them motivated every step of the way.

References

Mobile/Web Technologies & Frameworks

- ★ Google. Flutter Documentation. Available: **https://docs.flutter.dev/**
- ★ Google. *Firebase Documentation* (Cloud Firestore, Firebase Authentication, Realtime Database, Cloud Messaging). Available: **https://firebase.google.com/docs**
- ★ Google DeepMind. *Gemini API and Google AI Studio Documentation*. Available: **https://ai.google.dev/**
- ★ Dart Team. *Dart Programming Language Documentation*. Available: **https://dart.dev/**
- ★ PHP Group. *PHP Documentation*. Available: **https://www.php.net/docs.php**
- ★ Apache Friends. *XAMPP*. Available: **https://www.apachefriends.org/**
- ★ Microsoft. *Visual Studio Code*. Available: **https://code.visualstudio.com/**
- ★ GitHub, Inc. *GitHub*. Available: **https://github.com/**
- ★ Flutter Community. *Flutter Packages Documentation* (FL Chart, Flutter Local Notifications, Table Calendar, Percent Indicator, Lottie, Provider, Shared Preferences, URL Launcher, PDF, Printing, Intl). Available: **https://pub.dev/**

Related Works

[1] N. L. Inc., "Notion," [Online]. Available: **https://www.notion.so/.**

[2] D. Inc., "Todoist," Todoist, [Online]. Available: **https://todoist.com/.**

[3] A. Inc., "TickTick," TickTick, [Online]. Available: **https://ticktick.com/.**

[4] MyStudyLife, "MyStudyLife," MyStudyLife, [Online]. Available: **https://mystudylife.com/.**

[5] S. GmbH, "StudySmarter," StudySmarter, [Online]. Available: **https://www.studysmarter.co.uk/.**

[6] Seekrtech, "Forest," Forest, [Online]. Available: **https://www.forestapp.cc/.**

[7] SuperByte, "Study Bunny," Study Bunny, [Online]. Available: **https://superbyte.site/studybunny.**

[8] Motion, "Motion," Motion, [Online]. Available: **https://www.usemotion.com/.**

[9] Reclaim.ai, "Reclaim AI," Reclaim AI, [Online]. Available: **https://reclaim.ai/.**